Network Theory

Previous Year Questions

GATE ELECTRICAL ENGINEERING

(EE)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Network Theory (EE), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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- 5+ years of teaching experience

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

○○○ ●○○ Easy ●●○ Medium ●●● Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC **Questions:** 45 **MCQ:** 30 **MSQ:** 3 **NAT:** 12
1M: 20 **2M:** 22 **Descriptive:** 3 **Avg Weightage** ~4M **Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

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UNIT

I

Basic Concepts of Networks

Topics

1. Introduction to Electrical Quantities
2. Basic Laws and Classification of Network Elements
3. Resistors, Capacitors and Inductors
4. Energy Sources in Electrical Circuits
5. Series-Parallel connections
6. Methods of Analyzing Electrical Circuits - Nodal and Mesh Analysis

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Unit Snapshot*

Stream: EE	**Questions:** 42	**MCQ:** 26	**MSQ:** 0	**NAT:** 14
1M: 25	**2M:** 17	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2023)–3M(2010)

**See preface for how Avg Weightage and Range are calculated.*

Q1.26.1 MCQ 2026 1M

Refer to the four circuits shown.

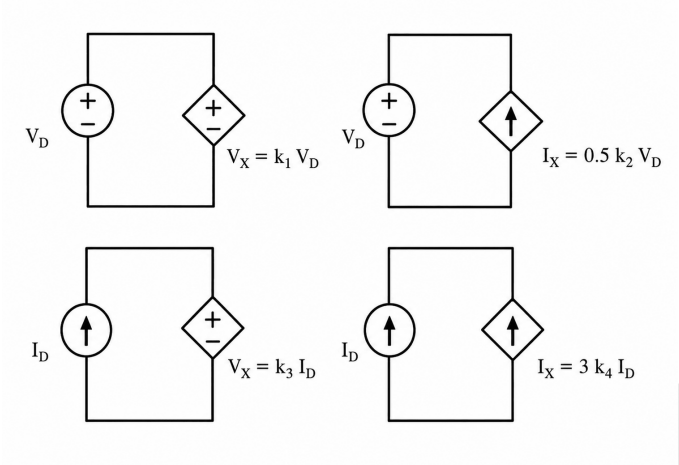


Fig. Q1.26.1

Which one of the following options for k_1, k_2, k_3, and k_4 makes all of them realizable?

A. $k_1 = 1, k_4 = -1/3$, for all values of k_2 and k_3

B. $k_2 = -2, k_3 = +1/3$, for all values of k_1 and k_4

C. $k_1 = 2, k_2 = 0.5, k_3 = -2/3, k_4 = -3$

D. $k_1 = 2, k_2 = -0.5, k_3 = -2/3, k_4 = +3$

Q1.26.2 MCQ 2026 1M

A circuit with ideal elements is shown.

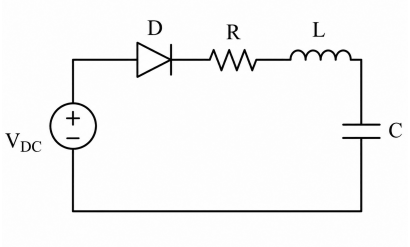


Fig. Q1.26.2

Which one of the following options correctly identifies all the linear elements in the circuit?

A. R only
B. R, L, and C only
C. D only
D. L, C, and D only

Q1.25.1 MCQ 2025 1M

A nullator is defined as a circuit element where the voltage across the device and the current through the device are both zero. A series combination of a nullator and a resistor of value, R, will behave as a

A. resistor of value R
B. nullator
C. open circuit
D. short circuit

Q1.25.2 MCQ 2025 1M

The I-V characteristics of the element between the nodes X and Y is best depicted by

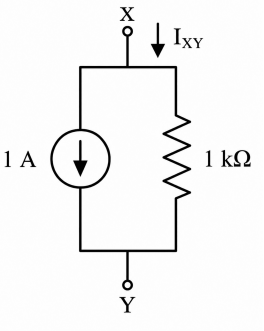
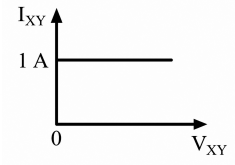
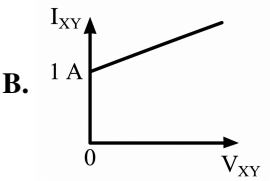
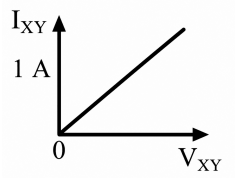
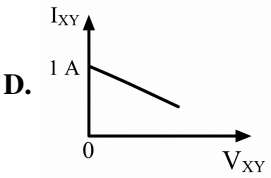


Fig. Q1.25.2

A.


B.


C.


D.


Q1.24.1 MCQ 2024 1M

The number of junctions in the circuit is

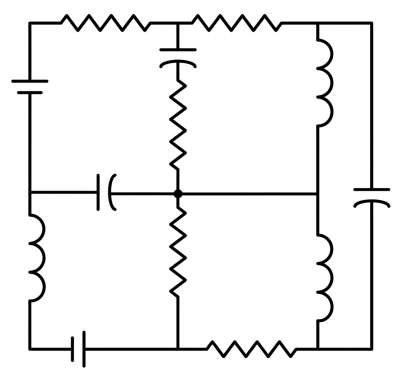


Fig. Q1.24.1

A. 6 **B.** 7 **C.** 8 **D.** 9

Q1.24.2 MCQ 2024 1M

All the elements in the circuit are ideal. The power delivered by the 10 V source in watts is

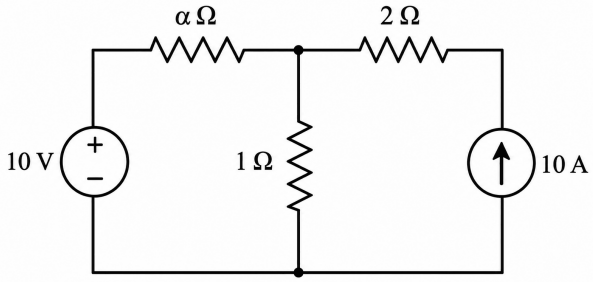


Fig. Q1.24.2

A. 0 **B.** 50
C. 100 **D.** dependent on the value of α

Q1.23.1 NAT 2023 1M

For the circuit shown in the figure, $V_1 = 8$ V, DC and $I_1 = 8$ A, DC. The voltage V_{ab} in Volts is _____(Round off to 1 decimal place).

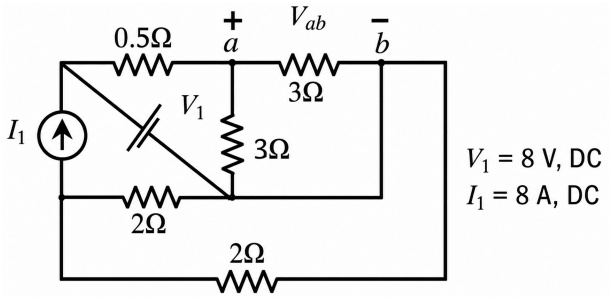


Fig. Q1.23.1

Q1.22.1 NAT 2022 2M

In the circuit shown below, the magnitude of voltage V_1 in volts, across the 8 kΩ resistor is _____. (round off to nearest integer)

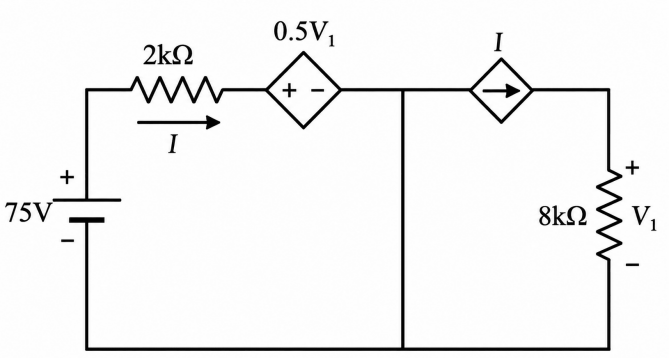


Fig. Q1.22.1

Q1.21.1 NAT 2021 1M

In the given circuit, for voltage V_y to be zero, value of β should be _____. (Round off to 2 decimal places).

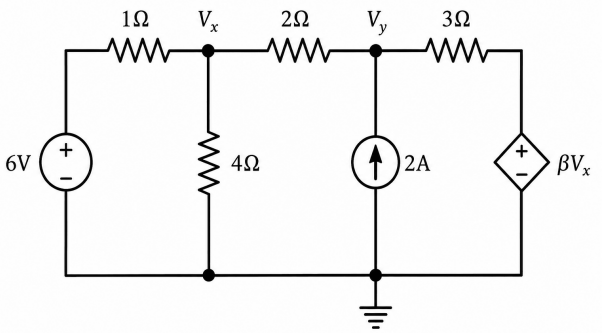


Fig. Q1.21.1

Q1.19.1 NAT 2019 1M

The current I flowing in the circuit shown below in amperes (round off to one decimal place) is _____.

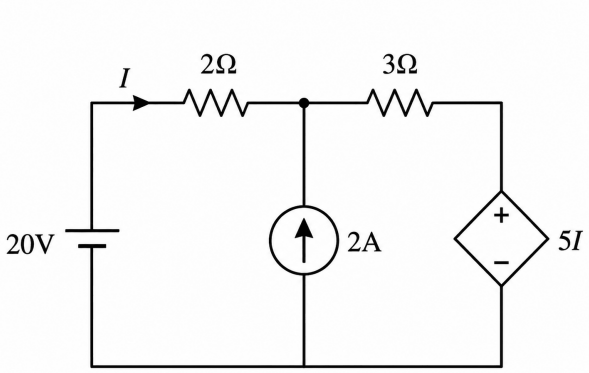


Fig. Q1.19.1

Q1.17.1 NAT 2017 1M

The power supplied by the 25 V source in the figure shown below is _____ W.

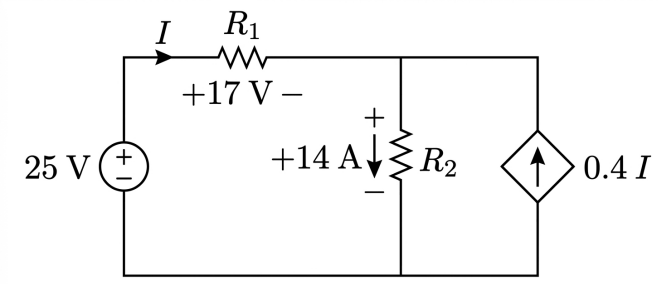


Fig. Q1.17.1

Q1.16.1 MCQ 2016 1M

In the circuit shown below, the voltage and current sources are ideal. The voltage (V_{out}) across the current source, in volts, is

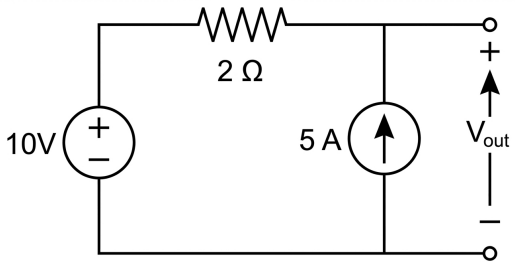


Fig. Q1.16.1

A. 0 **B.** 5 **C.** 10 **D.** 20

Q1.16.2 NAT 2016 1M

In the given circuit, the current supplied by the battery, in ampere, is _____.

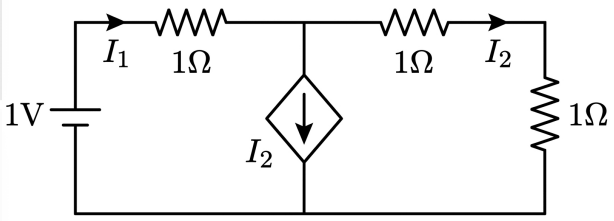


Fig. Q1.16.2

Q1.16.3 NAT 2016 1M

In the portion of a circuit shown, if the heat generated in 5 Ω resistance is 10 calories per second, then heat generated by the 4 Ω resistance, in calories per second, is _____.

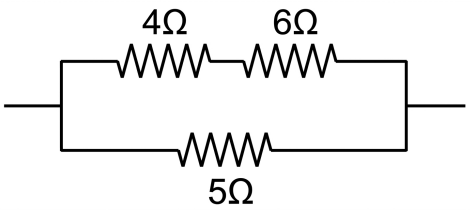


Fig. Q1.16.3

Q1.16.4 MCQ 2016 1M

The graph associated with an electrical network has 7 branches and 5 nodes. The number of independent KCL equations and the number of independent KVL equations, respectively, are

A. 2 and 5 **B.** 5 and 2 **C.** 3 and 4 **D.** 4 and 3

Q1.15.1 NAT 2015 2M

In the given circuit, the parameter k is positive, and the power dissipated in the 2 Ω resistor is 12.5W. The value of k is _____.

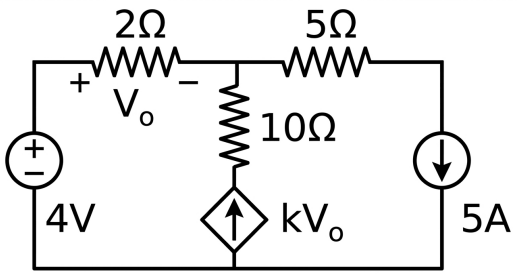


Fig. Q1.15.1

Q1.15.2 MCQ 2015 1M

The voltages developed across the 3 Ω and 2 Ω resistors shown in the figure are 6V and 2V respectively, with the polarity as marked. What is the power (in Watt) delivered by the 5V voltage source?

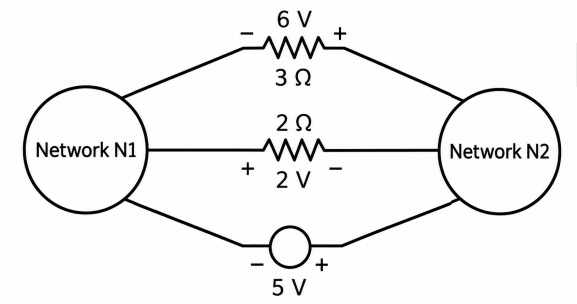


Fig. Q1.15.2

A. 5 **B.** 7 **C.** 10 **D.** 14

Q1.15.3 NAT 2015 1M

The current i (in Ampere) in the 2 Ω resistor of the given network is _____.

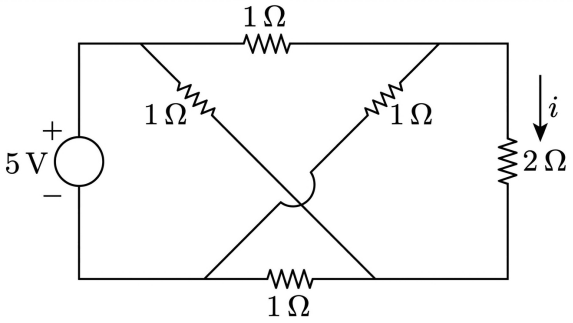


Fig. Q1.15.3

Q1.14.1 NAT 2014 2M

In the figure, the value of resistor R is $(25 + I/2)$ ohms, where I is the current in amperes. The current I is _____.

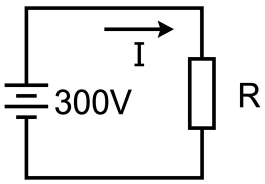


Fig. Q1.14.1

Q1.14.2 NAT 2014 2M

An incandescent lamp is marked 40W, 240V. If resistance at room temperature (26°C) is 120 Ω and temperature co-efficient of resistance is 4.5×10^{-3}/°C, then its 'ON' state filament temperature is °C is approximately _____.

Q1.14.3 NAT 2014 1M

The three circuit elements shown in the figure are part of an electric circuit. The total power absorbed by the three circuit elements in watt is _____.

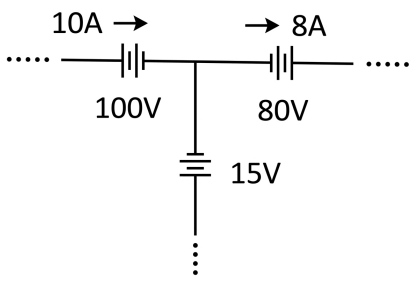


Fig. Q1.14.3

Q1.10.1 MCQ 2010 2M

If the 12 Ω resistor draws a current of 1 A as shown in the figure, the value of resistance R is

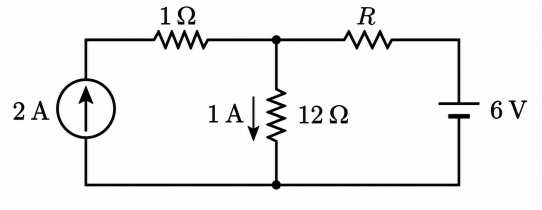


Fig. Q1.10.1

A. 4 Ω **B.** 6 Ω **C.** 8 Ω **D.** 18 Ω

Q1.10.2 MCQ 2010 1M

As shown in the figure given below, a 1 Ω resistance is connected across a source that has a load line $v + i = 100$. The current through the resistance is

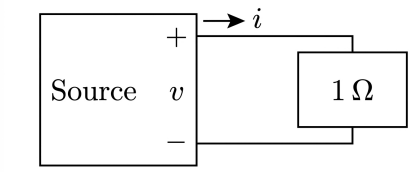


Fig. Q1.10.2

A. 25 A **B.** 50 A **C.** 100 A **D.** 200 A

Q1.09.1 MCQ 2009 2M

For the circuit shown below, find out the current flowing through the 2 Ω resistance. Also, identify the changes to be made to double the current through the 2 Ω resistance.

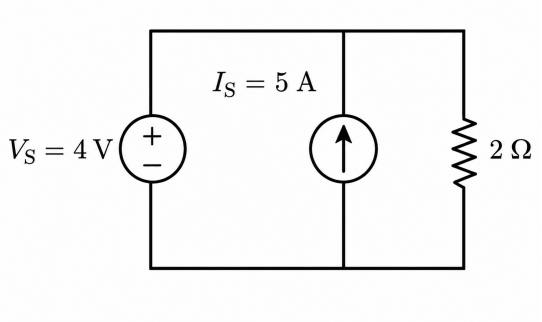


Fig. Q1.09.1

A. (5 A; Put $V_s = 20$ V) **B.** (2 A; Put $V_s = 8$ V)
C. (5 A; Put $I_s = 10$ A) **D.** (7 A; Put $I_s = 12$ A)

Q1.08.1 MCQ 2008 2M

In the circuit shown in the figure. The value of the current I will be given by

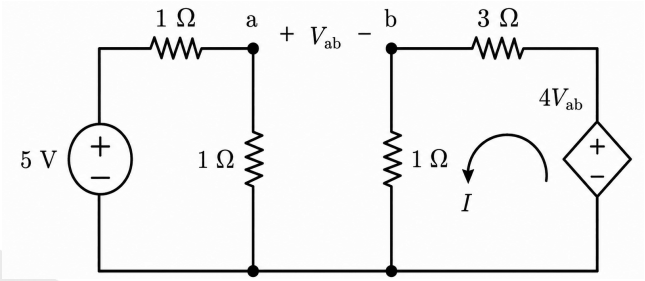


Fig. Q1.08.1

A. 0.31 A **B.** 1.25 A **C.** 1.75 A **D.** 2.5 A

Q1.08.2 MCQ 2008 2M

Assuming ideal elements in the circuit shown below, the voltage V_{ab} will be

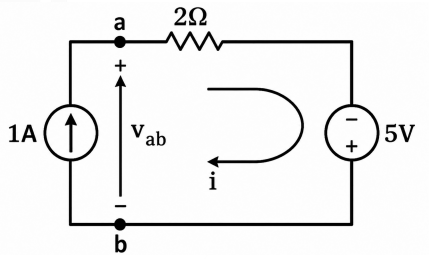


Fig. Q1.08.2

A. −3V **B.** 0V **C.** 3V **D.** 5V

Q1.07.1 MCQ 2007 2M

A 3 V dc supply with an internal resistance of 2 Ω supplies a passive non-linear resistance characterized by the relation $V_{NL} = I_{NL}^2$. The power dissipated in the non linear resistance is

A. 1.0 W **B.** 1.5 W **C.** 2.5 W **D.** 3.0 W

Q1.05.1 MCQ 2005 1M

In the given figure, the value of R is

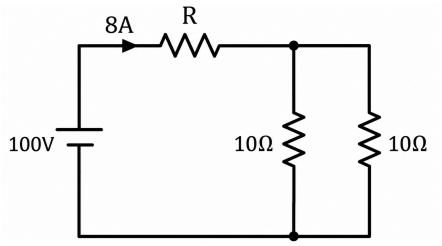


Fig. Q1.05.1

A. 2.5 Ω **B.** 5.0 Ω **C.** 7.5 Ω **D.** 10.0 Ω

Q1.04.1 MCQ 2004 2M

In the given figure, the value of the source voltage E is

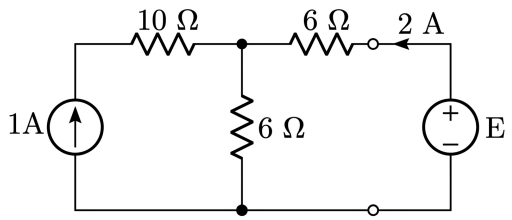


Fig. Q1.04.1

A. 12 V **B.** 24 V **C.** 30 V **D.** 44 V

Q1.03.1 MCQ 2003 1M

Figure shows the waveform of the current passing through an inductor of resistance 1 Ω and inductance 2 H. The energy absorbed by the inductor in the first four seconds is

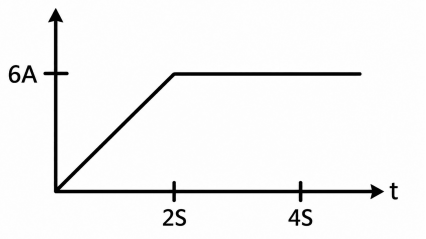


Fig. Q1.03.1

A. 144 J **B.** 98 J **C.** 132 J **D.** 168 J

Q1.01.1 MCQ 2001 2M

Consider the star network shown in Figure. The resistance between terminals A and B with C open is 6 Ω, between terminals B and C with A open is 11 Ω, and between terminals C and A with B open is 9 Ω. Then

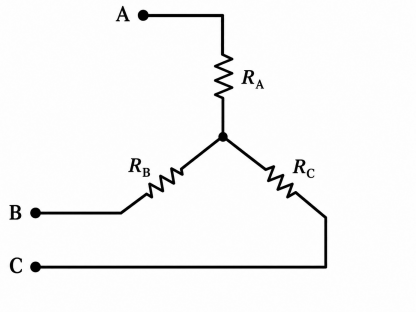


Fig. Q1.01.1

A. $R_A = 4\ \Omega, R_B = 2\ \Omega, R_C = 5\ \Omega$

B. $R_A = 2\ \Omega, R_B = 4\ \Omega, R_C = 7\ \Omega$

C. $R_A = 3\ \Omega, R_B = 3\ \Omega, R_C = 4\ \Omega$

D. $R_A = 5\ \Omega, R_B = 1\ \Omega, R_C = 10\ \Omega$

Q1.99.1 NAT 1999 2M

Solve the circuit shown in figure using the mesh method of analysis and determine the mesh currents I_1, I_2 and I_3. Evaluate the power developed in the 10V voltage source.

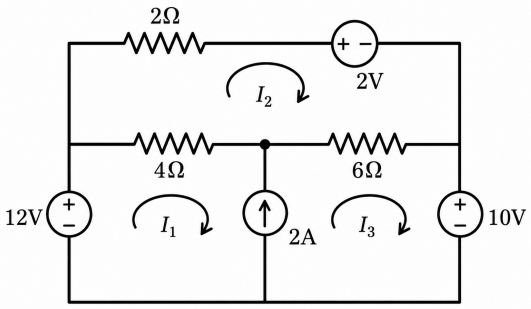


Fig. Q1.99.1

Q1.99.2 MCQ 1999 2M

When a resistor R is connected to a current source, it consumes a power of 18W. When the same R is connected to a voltage source having the same magnitude as the current source, the power absorbed by R is 4.5W. The magnitude of the current source and the value of R are

A. $\sqrt{18}$A and 1 Ω
B. 3A and 2 Ω
C. 1A and 18 Ω
D. 6A and 0.5 Ω

Q1.99.3 MCQ 1999 2M

The color code of a 1kΩ resistance is:

A. black, brown, red
B. red, brown, brown
C. brown, black, red
D. black, black, red

Q1.97.1 Descriptive 1997 2M

The voltage and current waveforms for an element are shown in figure. The circuit element is _____ and its value is _____.

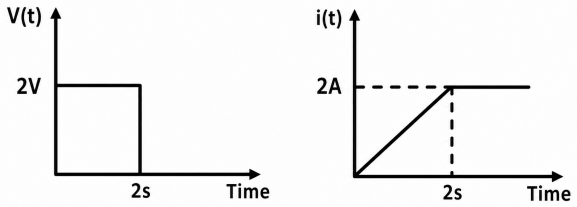


Fig. Q1.97.1

Q1.97.2 Descriptive 1997 2M

The values of E and I for the circuit shown in figure, are. . . . . V and . . . . . .

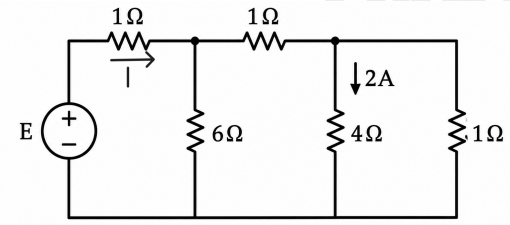


Fig. Q1.97.2

Q1.97.3 NAT 1997 2M

A 10V battery with an internal resistance of 1 Ω is connected across a nonlinear load whose V-I characteristic is given by $7I = V^2 + 2V$. The current delivered by the battery is _____A.

Q1.97.4 MCQ 1997 1M

Energy stored in a capacitor over a cycle, when excited by an a.c. source is

A. The same as that due to a d.c. source of equivalent magnitude
B. Half of that due to a d.c. source of equivalent magnitude
C. Zero
D. None of the above

Q1.97.5 MCQ 1997 1M

A practical current source is usually represented by

A. Resistance in series with an ideal current source
B. A resistance in parallel with an ideal current source
C. A resistance in parallel with an ideal voltage source
D. None of the above

Q1.97.6 MCQ 1997 1M

An ideal voltage source will charge an ideal capacitor

A. In infinite time
B. Exponentially
C. Instantaneously
D. None of these

Q1.96.1 MCQ 1996 1M

In the circuit shown in figure X is an element which always absorbs power. During a particular operations, it sets up a current of 1 amp in the direction shown and absorbs a power P_x. It is possible that X can absorb the same power P_x for another current i, the value of this current is

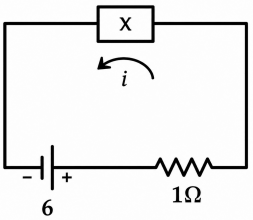


Fig. Q1.96.1

A. $(3 - \sqrt{14})$ amps
B. $(3 + \sqrt{14})$ amps
C. 5 amps
D. None of these

Q1.92.1 MCQ 1992 1M

All resistances in the circuit in figure are of R ohms each. The switch is initially open. What happens to the lamp's intensity when the switch is closed?

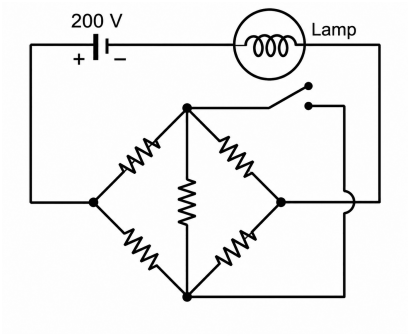


Fig. Q1.92.1

A. Increases **B.** Decreases
C. Remains same **D.** Answer depends on the value of R

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q1.26.1 A	Q1.26.2 B	Q1.25.1 B
Q1.25.2 B	Q1.24.1 A	Q1.24.2 A
Q1.23.1 5.9 to 6.1	Q1.22.1 98 to 102	Q1.21.1 -3.30 to -3.20
Q1.19.1 1.3 to 1.5	Q1.17.1 248 to 252	Q1.16.1 D
Q1.16.2 0.5 to 0.5	Q1.16.3 1.9 to 2.1	Q1.16.4 D
Q1.15.1 0.48 to 0.52	Q1.15.2 A	Q1.15.3 0
Q1.14.1 10 to 10	Q1.14.2 2469 to 2473	Q1.14.3 330 to 330
Q1.10.1 B	Q1.10.2 B	Q1.09.1 B
Q1.08.1 B	Q1.08.2 A	Q1.07.1 A
Q1.05.1 C	Q1.04.1 C	Q1.03.1 C
Q1.01.1 B	Q1.99.1 *	Q1.99.2 B
Q1.99.3 C	Q1.97.1 Inductor, 2H	Q1.97.2 31V, 13A
Q1.97.3 5	Q1.97.4 C	Q1.97.5 B
Q1.97.6 C	Q1.96.1 C	Q1.92.1 C

UNIT

II

Transient Analysis

Topics

1. Introduction to Transient Analysis
2. Basics of Signals for Transient Analysis
3. RC Circuits - Introduction and Examples
4. Impulse Current in Capacitor and Basics of Laplace in Circuit Analysis
5. Slope at $t=0^+$, Switched Capacitor and Impulse Response of RC Circuits
6. RC Circuits with Pulse Wave Input
7. Initial and Final Value Theorems
8. RL Circuits

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 50	**MCQ:** 37	**MSQ:** 0	**NAT:** 13
1M: 20	**2M:** 26	**Descriptive:** 4	**Avg Weightage** ~2M	**Range:** 1M(2024)–4M(2025)

**See preface for how Avg Weightage and Range are calculated.*

Q2.26.1 MCQ 2026 2M

The electrical network shown has an independent voltage source (10 V) and a current source (1 u(t) mA).

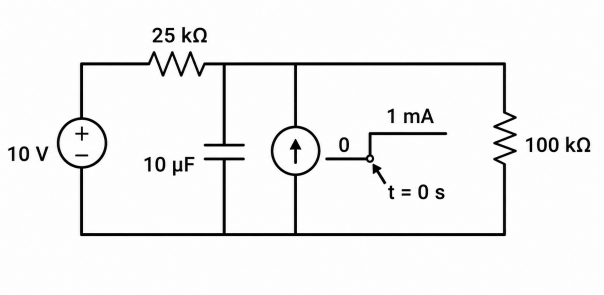


Fig. Q2.26.1

The voltage across the capacitor at time instants (in seconds) $t = 0^+$, $t = 0.50$, and $t = \infty$, respectively, is:

A. 8.00 V, 28.00 V, 26.36 V

B. 8.00 V, 26.36 V, 28.00 V

C. 10.00 V, 26.36 V, 28.00 V

D. 10.00 V, 28.00 V, 26.36 V

Q2.25.1 MCQ 2025 2M

In the circuit, shown below, if the values of R and C are very large, the form of the output voltage for a very high frequency square wave input, is best represented by

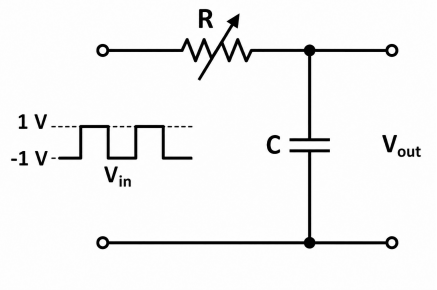
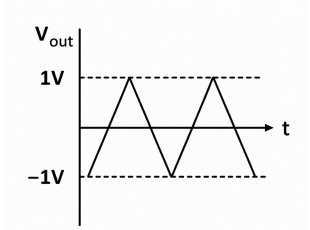
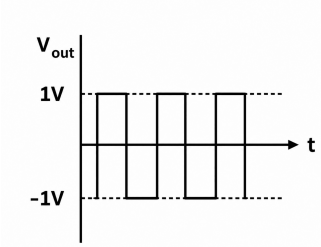
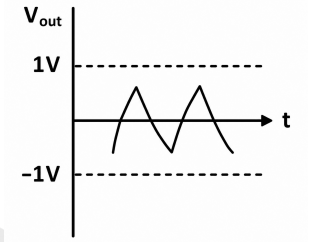
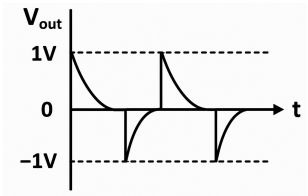


Fig. Q2.25.1

A.


B.


C.


D.


Q2.25.2 NAT 2025 2M

The switch (S) closes at t = 0sec. The time, in sec, the capacitor takes to charge to 50 V is _____ (round off to one decimal place).

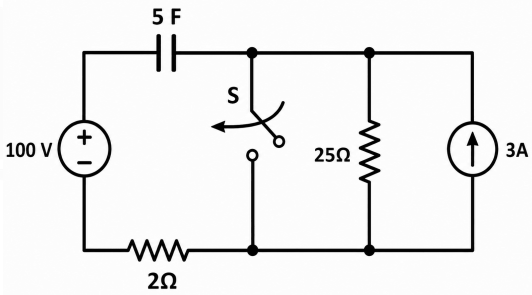


Fig. Q2.25.2

Q2.24.1 MCQ 2024 1M

The circuit shown in the figure with the switch S open, is in steady state. After the switch S is closed, the time constant of the circuit in seconds is

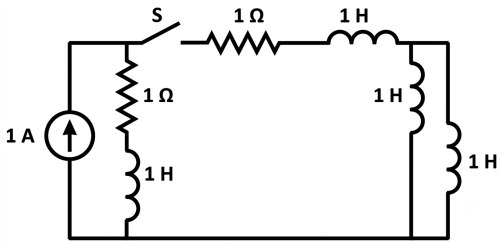


Fig. Q2.24.1

A. 1.25 **B.** 0 **C.** 1 **D.** 1.5

Q2.23.1 NAT 2023 1M

The value of parameters of the circuit shown in the figure are: $R_1 = 2\ \Omega, R_2 = 2\ \Omega, R_3 = 3\ \Omega, L = 10\ \text{mH}, C = 100\ \mu\text{F}$ For time $t < 0$, the circuit is at steady state with the switch 'K' in closed condition. If the switch is opened at $t = 0$, the value of the voltage across the inductor (V_L) at $t = 0^+$ in Volts is _____ (Round off to 1 decimal place).

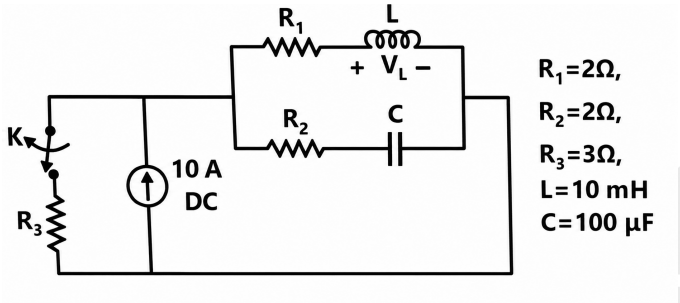


Fig. Q2.23.1

Q2.22.1 NAT 2022 1M

In the circuit shown below, the switch S is closed at $t = 0$. The magnitude of the steady state voltage, in volts, across the 6 Ω resistor is _____ (round off to two decimal places).

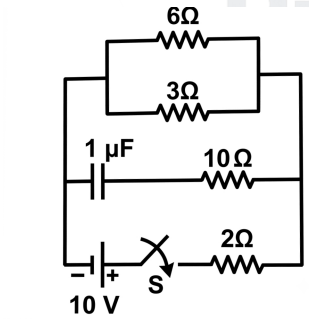


Fig. Q2.22.1

Q2.21.1 NAT 2021 2M

A 100 Hz square wave, switching between 0V and 5V, is applied to a CR high-pass filter circuit as shown. The output voltage waveform across the resistor is 6.2V peak-to-peak. If the resistance R is 820 Ω, then the value C is _____ μF. (Round off to 2 decimal places.)

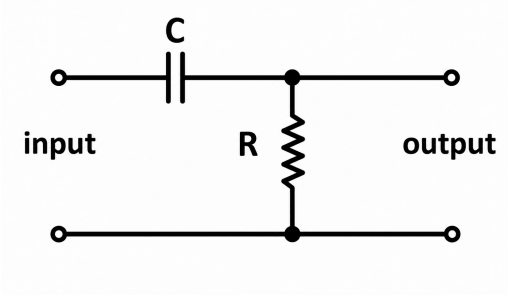


Fig. Q2.21.1

Q2.21.2 MCQ 2021 2M

In the circuit, switch 'S' is in the closed position for a very long time. If the switch is opened at time $t = 0$, then $i_L(t)$ in amperes, for $t \geq 0$ is

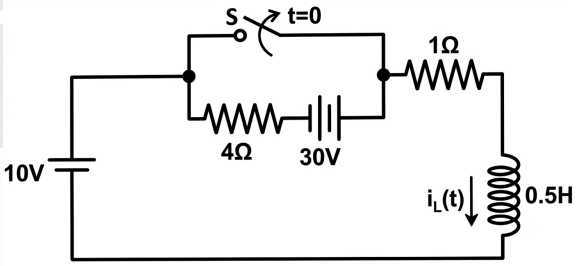


Fig. Q2.21.2

A. 10 **B.** $8\,e^{-10t}$ **C.** $8 + 2\,e^{-10t}$ **D.** $10(1 - e^{-2t})$

Q2.20.1 NAT 2020 2M

A resistor and a capacitor are connected in series to a 10 V DC supply through a switch. The switch is closed at $t = 0$, and the capacitor voltage is found to cross 0V at $t = 0.4\tau$, where τ is the circuit time constant. The absolute value of percentage change required in the initial capacitor voltage if the zero crossing has to happen at $t = 0.2\tau$ is _____ (rounded off to 2 decimal places).

Q2.19.1 NAT 2019 2M

A 0.1 μF capacitor charged to 100 V is discharged through a 1 kΩ resistor. The time in ms (round off to two decimal places) required for the voltage across the capacitor to drop to 1 V is _____

Q2.17.1 NAT 2017 1M

The initial charge in the 1 F capacitor present in the circuit shown is zero. The energy in joules transferred from the DC source until steady state condition is reached equals _____. (Give the answer up to one decimal place).

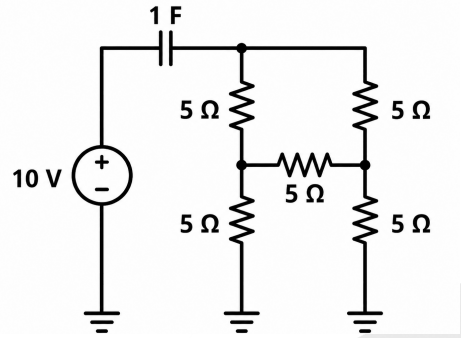


Fig. Q2.17.1

Q2.17.2 MCQ 2017 2M

The switch in the figure below was closed for a long time. It is opened at $t = 0$. The current in the inductor of 2 H for $t \geq 0$, is

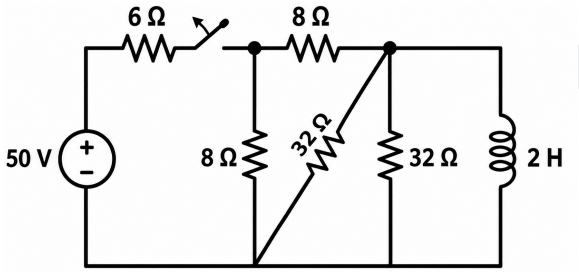


Fig. Q2.17.2

A. $2.5e^{-4t}$ **B.** $5e^{-4t}$ **C.** $2.5e^{-0.25t}$ **D.** $5e^{-0.25t}$

Q2.16.1 NAT 2016 2M

In the circuit shown below, the initial capacitor voltage is 4V, Switch S_1 is closed at $t = 0$. The charge (in μC) lost by the capacitor from $t = 25$ micro sec. to $t = 100$ μs is _____.

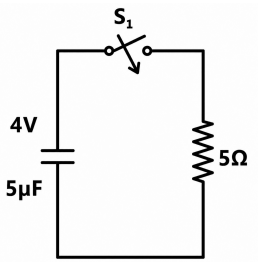


Fig. Q2.16.1

Q2.16.2 NAT 2016 2M

In the circuit shown, switch S_2 has been closed for a long time. At time $t = 0$ switch S_1 is closed. At $t = 0^+$, the rate of change of current through the inductor, in amperes per second, is _____.

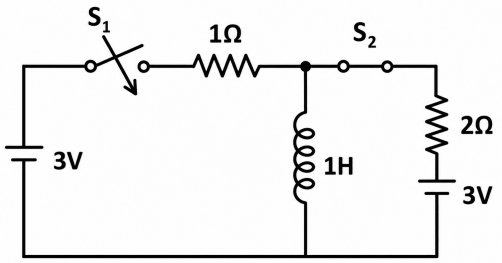


Fig. Q2.16.2

Q2.15.1 MCQ 2015 1M

A series RL circuit is excited at $t = 0$ by closing a switch as shown in the figure. Assuming zero initial conditions, the value of $\frac{d^2i}{dt^2}$ at $t = 0^+$ is

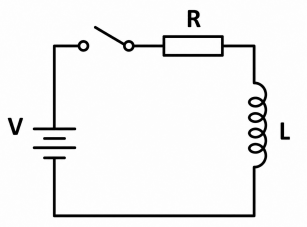


Fig. Q2.15.1

A. $\frac{V}{L}$ **B.** $\frac{-V}{R}$ **C.** 0 **D.** $\frac{-RV}{L^2}$

Q2.14.1 MCQ 2014 1M

The driving point impedance $Z(s)$ for the circuit shown below is

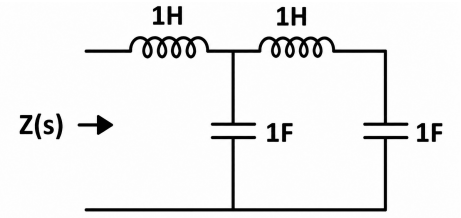


Fig. Q2.14.1

A. $\dfrac{s^4 + 3s^2 + 1}{s^3 + 2s}$

B. $\dfrac{s^4 + 2s^2 + 4}{s^2 + 2}$

C. $\dfrac{s^2 + 1}{s^4 + s^2 + 1}$

D. $\dfrac{s^3 + 1}{s^4 + s^2 + 1}$

Q2.14.2 MCQ 2014 1M

The switch SW shown in the circuit is kept at position '1' for a long duration. At $t = 0^+$, the switch is moved to position '2'. Assuming $|V_{02}| > |V_{01}|$, the voltage $V_C(t)$ across the capacitor is

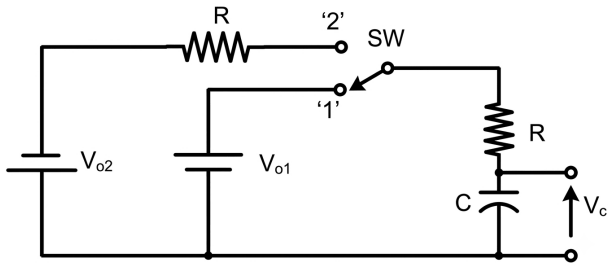


Fig. Q2.14.2

A. $V_C(t) = -V_{02}\left(1 - e^{-t/2RC}\right) - V_{01}$

B. $V_C(t) = V_{02}\left(1 - e^{-t/2RC}\right) + V_{01}$

C. $V_C(t) = -\left(V_{02} + V_{01}\right)\left(1 - e^{-t/2RC}\right) - V_{01}$

D. $V_C(t) = \left(V_{02} - V_{01}\right)\left(1 - e^{-t/2RC}\right) + V_{01}$

Q2.14.3 MCQ 2014 1M

A combination of 1 μF capacitor with an initial voltage $v_C(0) = -2$V in series with a 100 Ω resistor is connected to a 20mA ideal dc current source by operating both switches at $t = 0$ s as shown. Which of the following graphs shown in the options approximates the voltage V_S across the current source over the next few seconds?

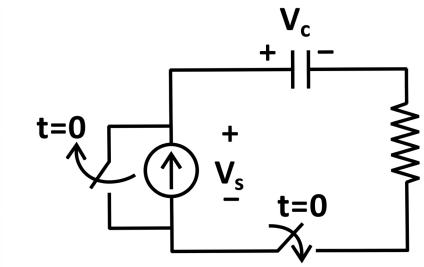
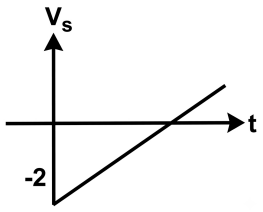
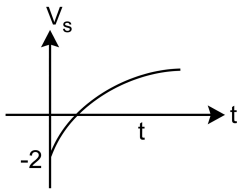
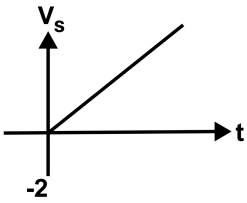
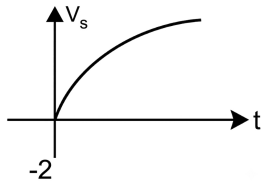


Fig. Q2.14.3

A.


B.


C.


D.


Q2.12.1 MCQ 2012 1M

In the following figure, C_1 and C_2 are ideal capacitors. C_1 has been charged to 12 V before the ideal switch S is closed at $t = 0$. The current $i(t)$ for all t is

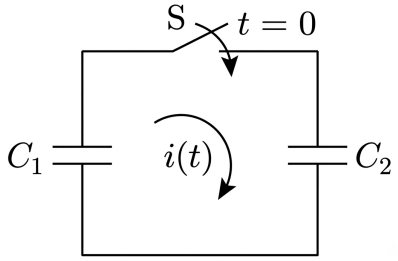


Fig. Q2.12.1

A. Zero

B. A step function

C. An exponentially decaying function

D. An impulse function

Q2.10.1 MCQ 2010 1M

The switch in the circuit has been closed for a long time. It is opened at $t = 0$. At $t = 0^+$, the current through the 1 μF capacitor is

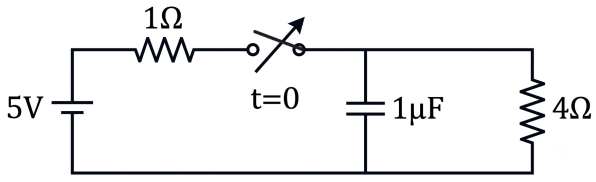


Fig. Q2.10.1

A. 0 A **B.** 1 A **C.** 1.25 A **D.** 5 A

Q2.09.1 MCQ 2009 2M

In the figure shown, all elements used are ideal. For time $t < 0$, S_1 remained closed and S_2 open. At $t = 0$, S_1 is opened and S_2 is closed. If the voltage V_{c2} across the capacitor C_2 at $t = 0$ is zero, the voltage across the capacitor combination at $t = 0^+$ will be

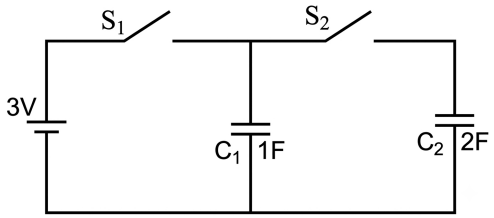


Fig. Q2.09.1

A. 1V **B.** 2V **C.** 1.5V **D.** 3V

Q2.09.2 MCQ 2009 1M

The equivalent capacitance of the input loop of the circuit shown below is

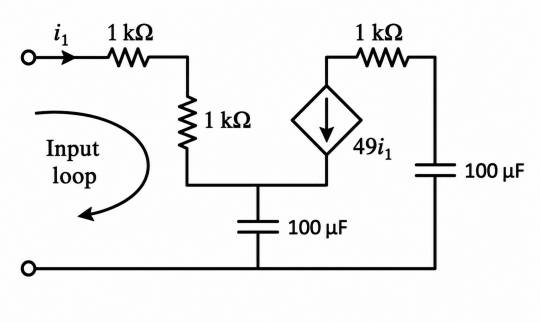


Fig. Q2.09.2

A. 2 μF **B.** 100 μF **C.** 200 μF **D.** 4 μF

Q2.08.1 MCQ 2008 2M

The time constant for the given circuit will be

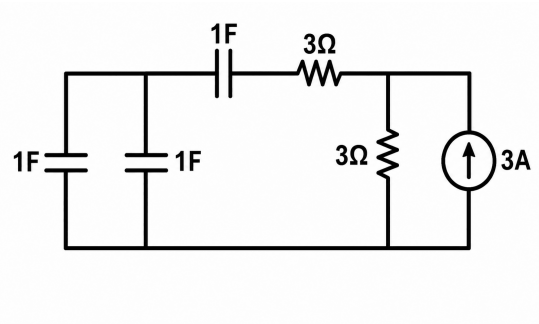


Fig. Q2.08.1

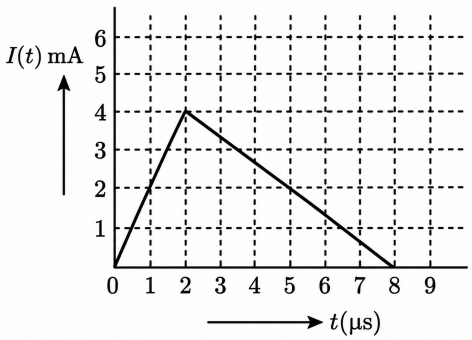
A. $\frac{1}{9}$ s **B.** $\frac{1}{4}$ s **C.** 4 s **D.** 9 s

Common Data Questions

The next 2 questions share the data below.

Statement for Linked Answer Questions Q2.08.2 and Q2.08.3

The current $I(t)$ sketched in the figure flows through an initially uncharged 0.3 nF capacitor.



Q2.08.2 MCQ 2008 2M

The charge stored in the capacitor at $t = 5$ μs, will be

A. 8 nC **B.** 10 nC **C.** 13 nC **D.** 16 nC

Q2.08.3 MCQ 2008 1M

The capacitor charged up to 5 μs, as per the current profile given in the figure, is connected across an inductor of 0.6 mH. Then value of voltage across the capacitor after 1 μs will approximately be

A. 18.8 V **B.** 23.5 V **C.** −23.5 V **D.** −30.6 V

Q2.07.1 MCQ 2007 2M

In the circuit shown in figure switch SW_1 is initially CLOSED and SW_2 is OPEN. The inductor L carries a current of 10A and the capacitor is charged to 10 V with polarities as indicated. SW_2, is initially CLOSED at $t = 0^-$ and SW_1, is OPENED at $t = 0$. The current through C and the voltage across L at $t = 0^+$ is

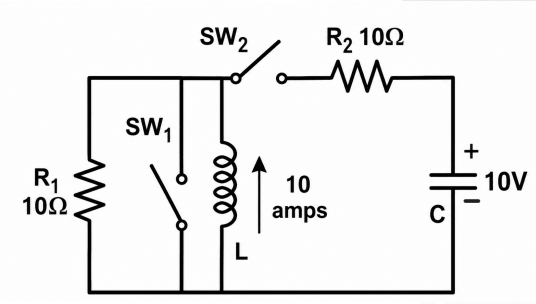


Fig. Q2.07.1

A. 55 A, 4.5 V **B.** 5.5 A, 4.5 V
C. 45 A, 5.5 V **D.** 4.5 A, 55 V

Q2.06.1 MCQ 2006 2M

In the circuit shown in the figure given below, current source $I = 1$ A, voltage source $V = 5$ V, $R_1 = R_2 = R_3 = 1\ \Omega$, $L_1 = L_2 = L_3 = 1$ H, $C_1 = C_2 = 1$ F. The currents (in A) through R_3 and the voltage source V, respectively, will be

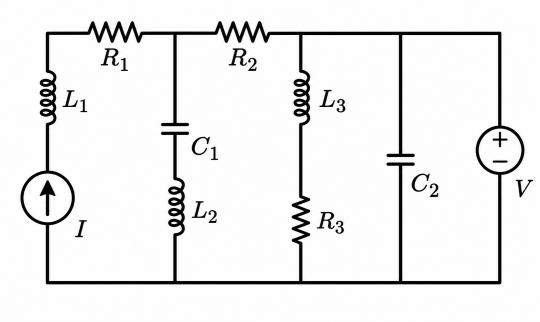


Fig. Q2.06.1

A. 1, 4 **B.** 5, 1 **C.** 5, 2 **D.** 5, 4

Q2.05.1 MCQ 2005 1M

In Figure, the initial capacitor voltage is zero. The switch is closed at $t = 0$. The final steady-state voltage across the capacitor is:

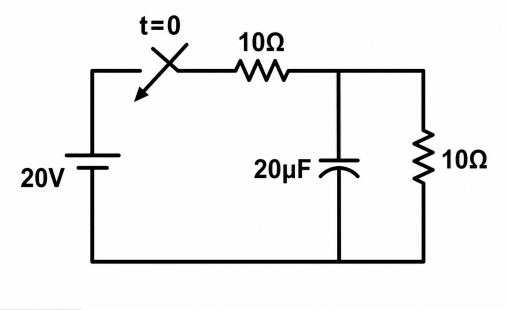


Fig. Q2.05.1

A. 20V **B.** 10V **C.** 5V **D.** 0V

Common Data Questions

The next 2 questions share the data below.

Statement for linked answer question Q2.05.2 and Q2.05.3 :

A coil of inductance 10H resistance 40 Ω is connected as shown in figure. After the switch S has been in connected with point 1 for a very long time, it is moved to point 2 at $t = 0$.

Q2.05.2 MCQ 2005 2M

If, at $t = 0+$, the voltage across the coil is 120V, the value of resistance R is:

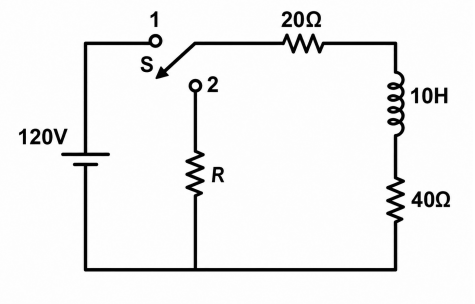


Fig. Q2.05.2

A. 0 Ω **B.** 20 Ω **C.** 40 Ω **D.** 60 Ω

Q2.05.3 MCQ 2005 2M

For the value of R obtained in Q2.05.2, the time taken for 95% of the stored energy dissipated is close to

A. 0.10 sec **B.** 0.15 sec **C.** 0.50 sec **D.** 1.0 sec

Q2.05.4 MCQ 2005 2M

The circuit shown in the figure is in steady state, when the switch is closed at $t = 0$. Assuming that the inductance is ideal, the current through the inductor at $t = 0^+$ equals

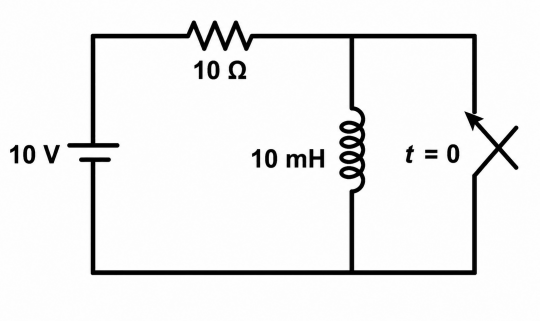


Fig. Q2.05.4

A. 0 A **B.** 0.5 A **C.** 1 A **D.** 2 A

Q2.04.1 MCQ 2004 2M

In figure, the capacitor initially has a charge of 10 Coulomb. The current in the circuit one second after the switch S is closed will be

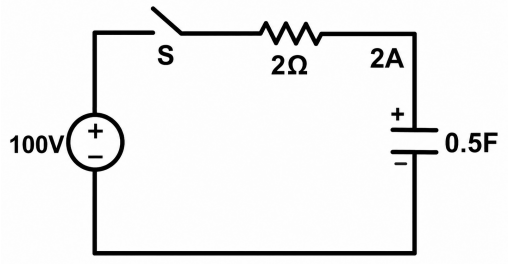


Fig. Q2.04.1

A. 14.7 A **B.** 18.5 A **C.** 40.0 A **D.** 50.0 A

Q2.03.1 MCQ 2003 2M

In the circuit shown in Figure, the switch S is closed at time $t = 0$. The voltage across the inductance at $t = 0^+$, is

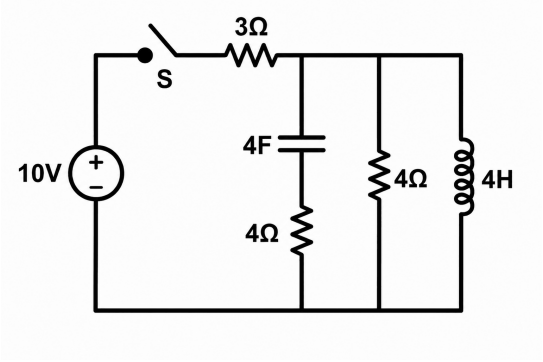


Fig. Q2.03.1

A. 2V **B.** 4V **C.** −6V **D.** 8V

Q2.02.1 NAT 2002 5M

A constant current source is supplying 10A to a circuit shown in Figure. The switch S, which is initially closed for a sufficiently long time, is suddenly opened. Obtain the differential equation governing the behavior of the inductor current and hence obtain the complete time response of the inductor current. What is the energy stored in L, a long time after the switch is opened?

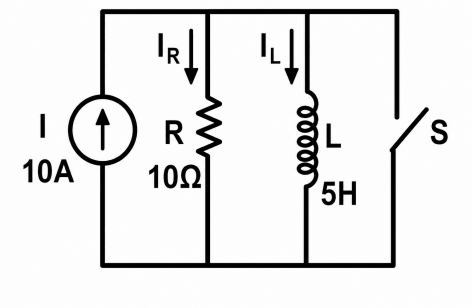


Fig. Q2.02.1

Q2.02.2 MCQ 2002 2M

In the circuit shown in Figure, the switch is closed at time $t = 0$. The steady state value of the voltage V_C is

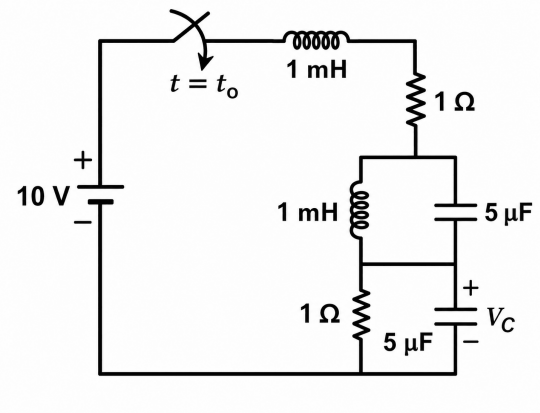


Fig. Q2.02.2

A. 0 V **B.** 10 V **C.** 5 V **D.** 2.5 V

Q2.02.3 MCQ 2002 2M

An 11 V pulse of 10 μs duration is applied to the circuit shown in Figure. Assuming that the capacitor is completely discharged prior to applying the pulse, the peak value of the capacitor voltage is

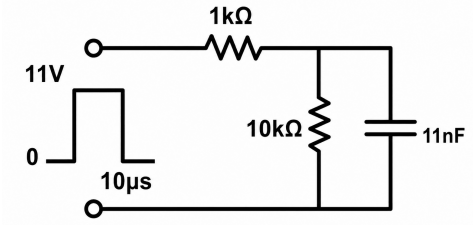


Fig. Q2.02.3

A. 11 V **B.** 5.5 V **C.** 6.32 V **D.** 0.96 V

Q2.02.4 MCQ 2002 1M

A current impulse, $5\delta(t)$, is forced through a capacitor C. The voltage, $V_C(t)$, across the capacitor is given by

A. $5t$ **B.** $5u(t) - C$ **C.** $\frac{5}{C}t$ **D.** $\frac{5u(t)}{C}$

Q2.01.1 MCQ 2001 1M

A unit step voltage is applied at $t = 0$ to a series RL circuit with zero initial conditions.

A. It is possible for the current to be oscillatory.

B. The voltage across the resistor at $t = 0^+$ is zero.

C. The energy stored in the inductor in the steady state is zero.

D. The resistor current eventually falls to zero.

Q2.00.1 MCQ 2000 1M

A voltage waveform $V(t) = 12t^2$ is applied across a 1H inductor for $t \geq 0$, with initial current through it being zero. The current through the inductor for $t \geq 0$ is given by

A. $12t$ **B.** $24t$ **C.** $12t^3$ **D.** $4t^3$

Q2.99.1 NAT 1999 5M

In the circuit shown in figure, capacitor is initially charged to 12V. Find the mathematical expression for the voltage across the capacitor V_c after closing the switch at $t = 0$

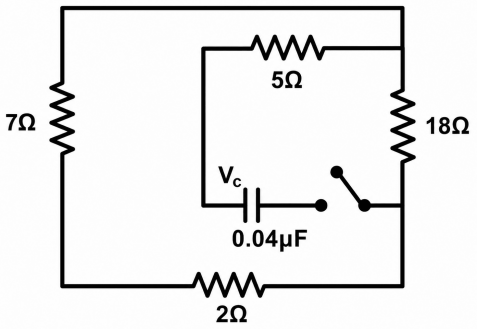
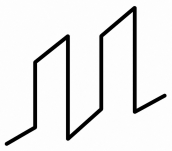


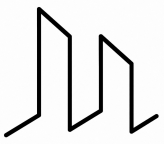
Fig. Q2.99.1

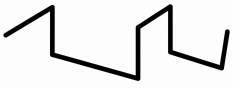
Q2.99.2 MCQ 1999 2M

When a periodic triangular voltage peak amplitude 1V and frequency 0.5Hz is applied to a parallel combination of 1 Ω resistance and 1F capacitance, the current through the voltage source has wave-form;

A. 

B.

C. 

D. 

Q2.99.3 MCQ 1999 2M

A rectangular voltage pulse of magnitude V and duration T is applied to a series combination of resistance R and capacitance C. The maximum voltage developed across the capacitor is:

A. $V\left[1-\exp\left(\frac{-T}{RC}\right)\right]$ **B.** $\frac{VT}{RC}$

C. V **D.** $V\exp\left(\frac{-T}{RC}\right)$

Q2.98.1 MCQ 1998 1M

In the circuit shown in figure, it is desired to have a constant direct current $i(t)$ through the ideal inductor L. The nature of the voltage source $v(t)$ must be:

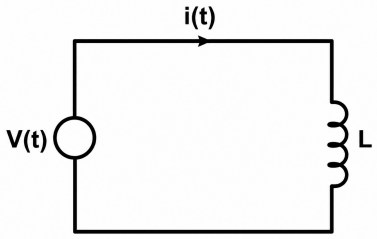


Fig. Q2.98.1

A. Constant voltage

B. Linearly increasing voltage

C. An ideal impulse

D. Exponentially increasing voltage

Q2.98.2 NAT 1998 5M

The switch in the following circuit, shown in fig 3, has been connected to the 12 V source for a long time. At $t = 0$, the switch is thrown to 24V. The value of L = 2H, $R_1 = 10\ \Omega$, $R_2 = 2\ \Omega$, C = 0.25 μF

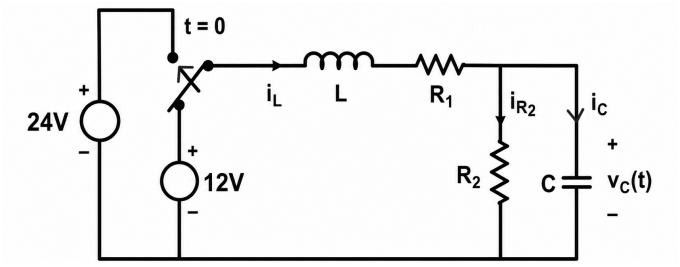


Fig. Q2.98.2

a) Determine $i_L(0)$ and $V_c(0)$

b) Write the differential equation governing $V_c(t)$ for $t > 0$

c) Compute the steady state value of $V_c(t)$

Q2.97.1 NAT 1997 5M

In the circuit shown in figure. $e_g(t) = 2.5t$ volts. What are the values of $i(t)$ and $V_L(t)$ at $t = 4$ seconds?

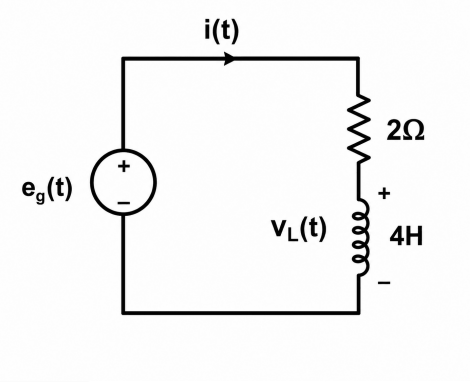


Fig. Q2.97.1

Q2.96.1 MCQ 1996 1M

In the series RC circuit shown in figure the voltage across C starts increasing when the d.c. source is switched on. The rate of increase of voltage across C at the instant just after the switch is closed (i.e., at $t = 0^+$), will be

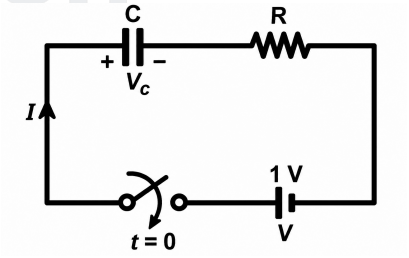


Fig. Q2.96.1

A. zero **B.** infinity **C.** RC **D.** 1/RC

Q2.96.2 MCQ 1996 1M

The v-i characteristic as seen from the terminal pair (A, B) of the network of Fig. 1 is shown in the Fig. (2). If an inductance of value 6 mH is connected across the terminal – pair (A, B), the time constant of the system will be

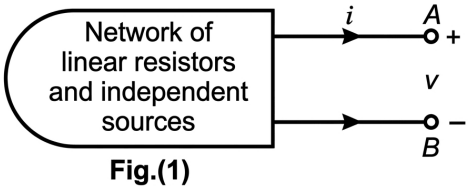


Fig.(1)

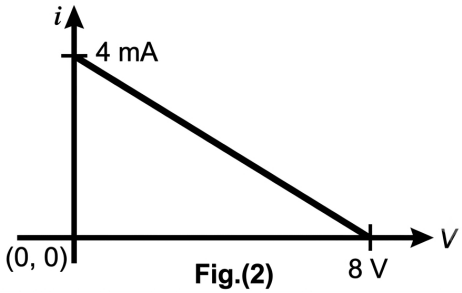


Fig.(2)

Fig. Q2.96.2

A. 3 μsec

B. 12 μsec

C. 32 μsec

D. unknown, unless the actual network is specified

Q2.92.1 MCQ 1992 1M

The time constant of the network shown in figure, is

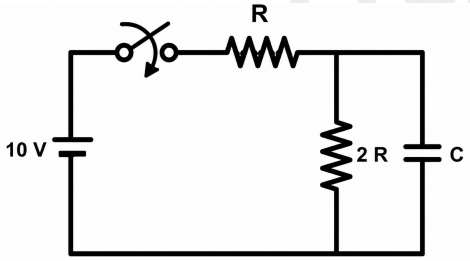


Fig. Q2.92.1

A. 2RC **B.** 3RC **C.** $\frac{RC}{2}$ **D.** $\frac{2RC}{3}$

Q2.91.1 MCQ 1991 2M

The switch S in figure is closed at $t = 0$. If $v_2(0) = 10\text{V}$ and $v_s(0) = 0\text{V}$ respectively, voltages across the capacitors in steady state will be :

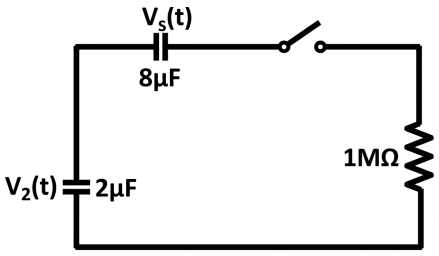


Fig. Q2.91.1

A. $v_2(\infty) = v_s(\infty) = 0$ **B.** $v_2(\infty) = 2\text{V}, v_s(\infty) = 8\text{V}$
C. $v_2(\infty) = v_s(\infty) = 8\text{V}$ **D.** $v_2(\infty) = v_s(\infty) = 2\text{V}$

Q2.91.2 MCQ 1991 2M

Two coils having equal resistance but different inductance are connected in series. The time-constant of the series combination is :

A. Sum of the time-constants of the individual coils

B. Average of the time constant of the individual coils

C. Geometric mean of the time constants of the individual coils

D. Product of the time-constants of the individual coils

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.
MTA Marks To All awarded — see detailed solutions.

Q2.26.1 B	Q2.25.1 C	Q2.25.2 4.0 to 4.2
Q2.24.1 A	Q2.23.1 7.9 to 8.1	Q2.22.1 4.95 to 5.05
Q2.21.1 12.30 to 12.60	Q2.21.2 C	Q2.20.1 54.00 to 56.00
Q2.19.1 0.45 to 0.47	Q2.17.1 99.0 to 101.0	Q2.17.2 A
Q2.16.1 6.8 to 7.2	Q2.16.2 1.9 to 2.1	Q2.15.1 D
Q2.14.1 A	Q2.14.2 MTA	Q2.14.3 C

Q2.12.1 D	Q2.10.1 B	Q2.09.1 A
Q2.09.2 A	Q2.08.1 C	Q2.08.2 C
Q2.08.3 D	Q2.07.1 D	Q2.06.1 D
Q2.05.1 B	Q2.05.2 C	Q2.05.3 B
Q2.05.4 C	Q2.04.1 A	Q2.03.1 B
Q2.02.1 *	Q2.02.2 C	Q2.02.3 C
Q2.02.4 D	Q2.01.1 B	Q2.00.1 D
Q2.99.1 *	Q2.99.2 D	Q2.99.3 A
Q2.98.1 C	Q2.98.2 *	Q2.97.1 *
Q2.96.1 D	Q2.96.2 A	Q2.92.1 D
Q2.91.1 D	Q2.91.2 B	

PrepFusion

UNIT

III

Network Theorems For DC & Miscellaneous Topics

Topics

1. Source Transformation and Superposition Theorem
2. Thevenin's Theorem
3. Norton's Theorem
4. Concept of Linearity
5. Reciprocity Theorem
6. Maximum Power Transfer Theorem
7. Star-Delta Connections and conversions
8. Finding Equivalent Resistances
9. Equivalent Resistance of a Cube Across Different Nodes
10. Equivalent Resistance in an Infinite Ladder
11. Bulb-Based Problems

Unit Snapshot*

Stream: EE	**Questions:** 36	**MCQ:** 23	**MSQ:** 0	**NAT:** 10
1M: 15	**2M:** 18	**Descriptive:** 3	**Avg Weightage** ~2M	**Range:** 1M(2021)–5M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Q3.26.1 MCQ 2026 1M

For the circuit shown, which one of the following options correctly identifies the Thevenin's equivalent parameters between nodes Y and Z ?

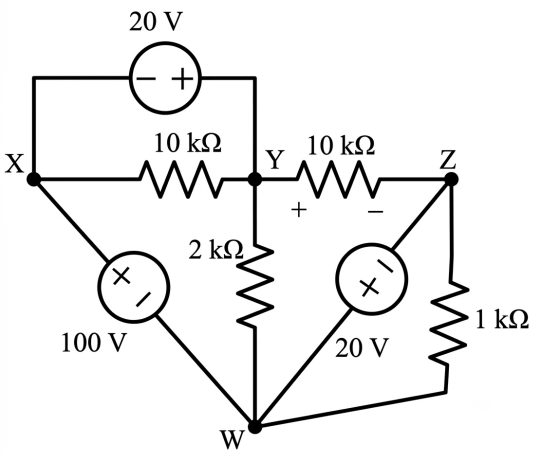


Fig. Q3.26.1

A. V_{TH} = 100 V, R_{TH} = 10kΩ

B. V_{TH} = 140 V, R_{TH} = 0kΩ

C. V_{TH} = 100 V, R_{TH} = 0kΩ

D. V_{TH} = 140 V, R_{TH} = 10kΩ

Q3.26.2 MCQ 2026 2M

The terminal voltage and current of a linear electrical network shown in Figure (a) are given in the table.

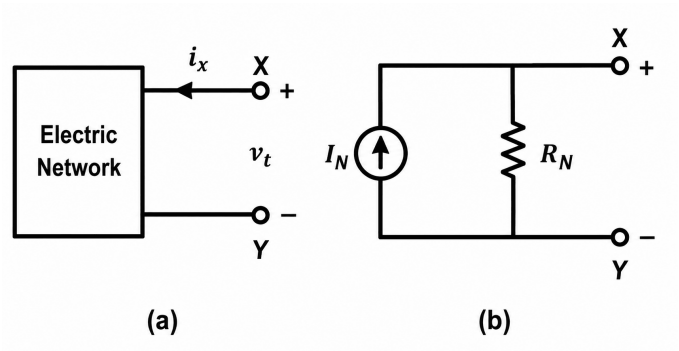


Fig. Q3.26.2

Terminal voltage (v_t)	Terminal current (i_x)
18 V	−0.5 A
30 V	0.5 A
36 V	1.0 A

The correct choice for the parameters (I_N, R_N) of the Norton equivalent circuit shown in Figure (b) is:

A. I_N = 3.0 A, R_N = 24.0 Ω

B. I_N = 12.0 A, R_N = 2.0 Ω

C. I_N = 2.0 A, R_N = 12.0 Ω

D. I_N = 2.0 A, R_N = 24.0 Ω

Q3.26.3 NAT 2026 2M

Consider the two-port network shown. For maximum power transfer to the resistive load (R_L), the value of R_L should be _____ Ω. (Round off to two decimal places)

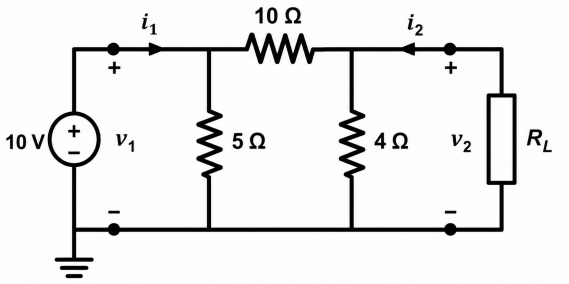


Fig. Q3.26.3

Q3.21.1 MCQ 2021 1M

For the network shown, the equivalent Thevenin voltage and Thevenin impedance as seen across terminals 'ab' is

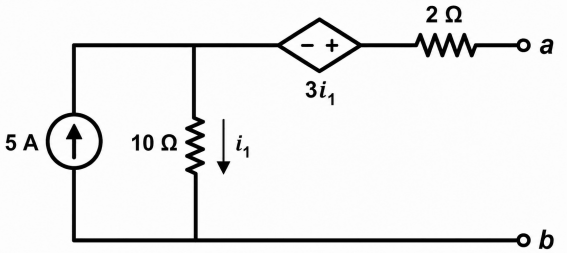


Fig. Q3.21.1

A. 10V in series with 12ohms

B. 65V in series with 15ohms

C. 50V in series with 2ohms

D. 35V in series with 2ohms

Q3.20.1 MCQ 2020 2M

A benchtop DC power supply acts as an ideal 4 A current source as long as its terminal voltage is below 10 V. Beyond this point, it begins to behave as an ideal 10 V voltage source for all load currents going down to 0A. When connected to an ideal rheostat, find the load resistance value at which maximum power is transferred, and the corresponding load voltage and current.

A. Short, ∞A, 10V
B. Open, 4 A, 0 V
C. 2.5 Ω, 4 A, 10V
D. 2.5 Ω, 4 A, 5V

Q3.20.2 NAT 2020 1M

The Thevenin equivalent voltage, V_{TH} in V (rounded off to 2 decimal places) of the network shown below, is

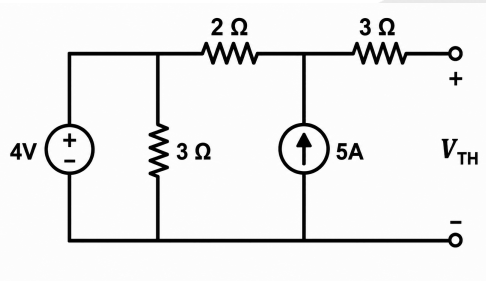


Fig. Q3.20.2

Q3.19.1 NAT 2019 2M

The current I flowing in the circuit shown below in amperes is _____.

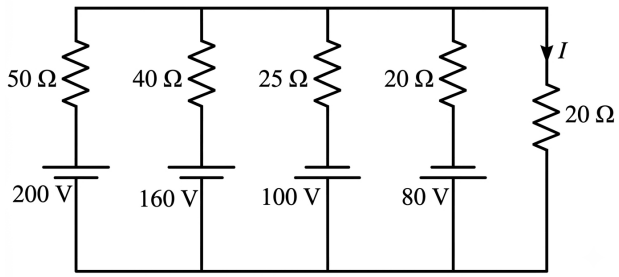


Fig. Q3.19.1

Q3.17.1 MCQ 2017 2M

For the network given in figure below, the Thevenin's voltage V_{ab} is

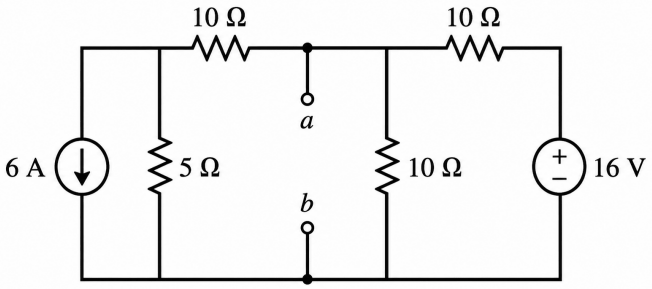


Fig. Q3.17.1

A. −1.5 V **B.** −0.5 V **C.** 0.5 V **D.** 1.5 V

Q3.17.2 NAT 2017 2M

In the circuit shown below, the maximum power transferred to the resistor R is _____ W.

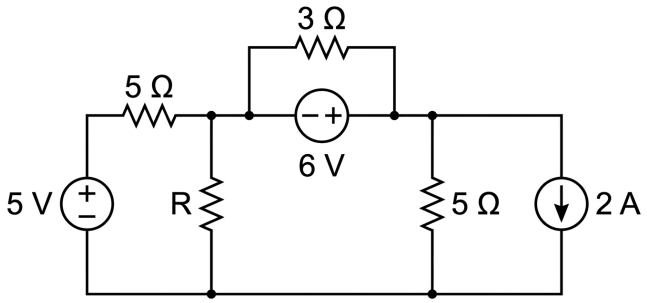


Fig. Q3.17.2

Q3.17.3 NAT 2017 1M

The equivalent resistance between the terminals A and B is _____ Ω.

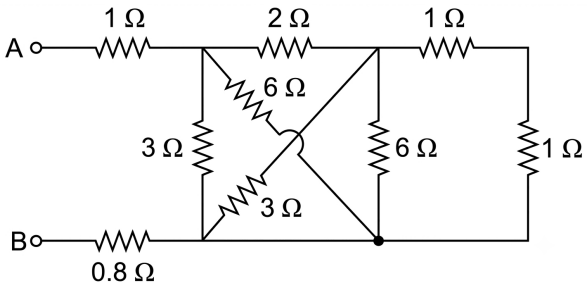


Fig. Q3.17.3

Q3.16.1 MCQ 2016 1M

R_A and R_B are the input resistances of circuits as shown below. The circuit extends infinitely in the direction shown. Which one of the following statements is TRUE?

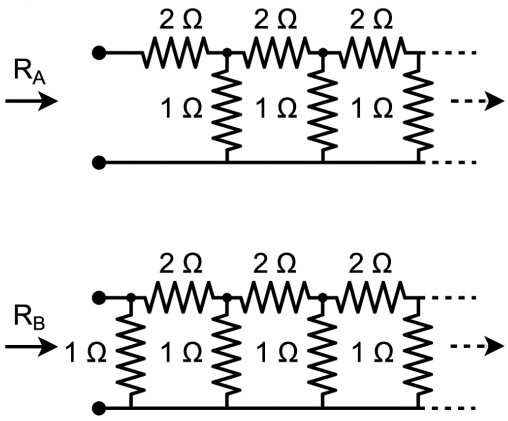


Fig. Q3.16.1

A. $R_A = R_B$
B. $R_A = R_B = 0$
C. $R_A < R_B$
D. $R_B = R_A/(1 + R_A)$

Q3.16.2 NAT 2016 2M

In the circuit shown below, the node voltage V_A is _____ V.

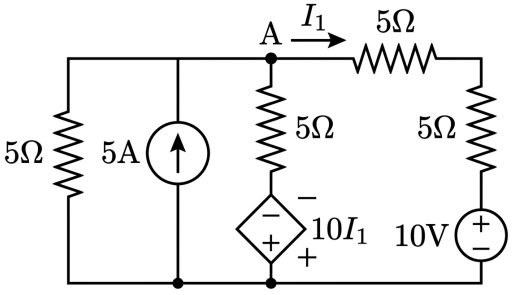


Fig. Q3.16.2

Q3.15.1 NAT 2015 1M

For the given circuit, the Thevenin equivalent is to be determined. The Thevenin voltage, V_{Th} (in Volt), seen from terminal AB is _____.

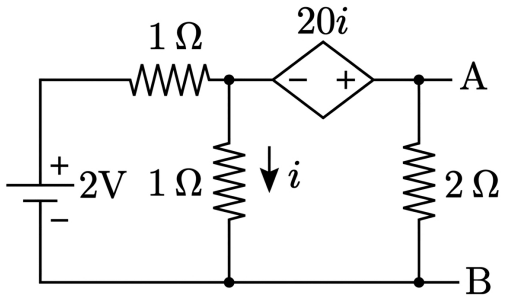


Fig. Q3.15.1

Q3.14.1 NAT 2014 2M

The power delivered by the current source, in the figure is _____.

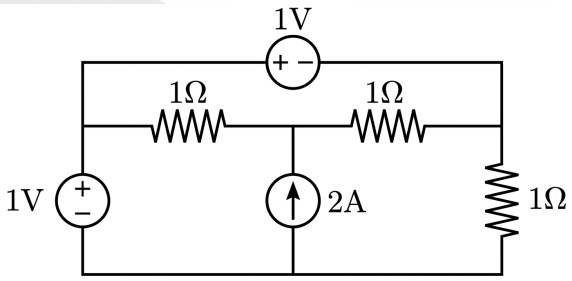


Fig. Q3.14.1

Q3.14.2 NAT 2014 2M

The Norton's equivalent source in amperes as seen into the terminals X and Y is _____.

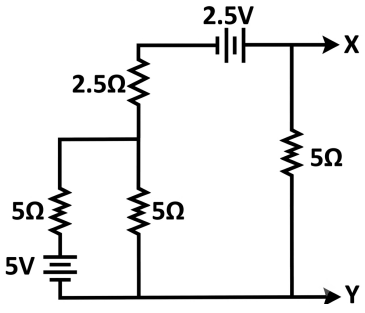


Fig. Q3.14.2

Q3.13.1 MCQ 2013 1M

Consider a delta connection of resistors and its equivalent star connection as shown below. If all elements of the delta connection are scaled by a factor k, $k > 0$, the element of the corresponding star equivalent will be scaled by a factor of

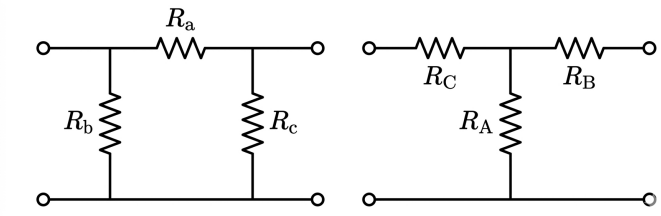


Fig. Q3.13.1

A. K^2 **B.** k **C.** $1/k$ **D.** $\sqrt{K}$

Q3.12.1 MCQ 2012 2M

If $V_A - V_B = 6$ V; then $V_C - V_D$ is

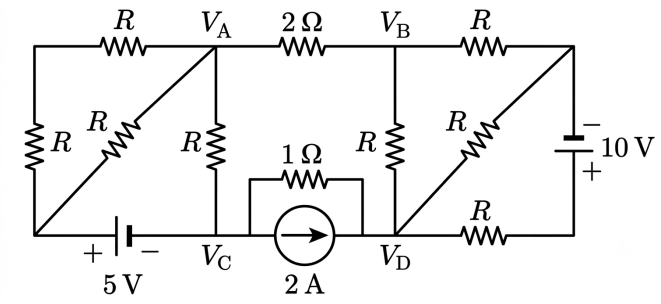


Fig. Q3.12.1

A. −5 V **B.** 2 V **C.** 3 V **D.** 6 V

Q3.12.2 MCQ 2012 1M

The impedance looking into nodes 1 and 2 in the given circuit is

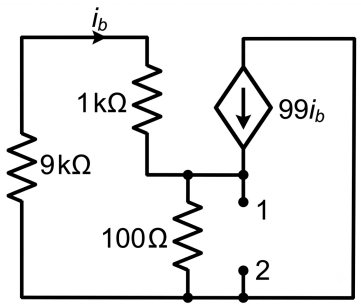


Fig. Q3.12.2

A. 50 Ω **B.** 100 Ω **C.** 5 kΩ **D.** 10.1 kΩ

Q3.11.1 MCQ 2011 1M

In the circuit given below, the value of R required for the transfer of maximum power to the load having a resistance of 3 Ω is

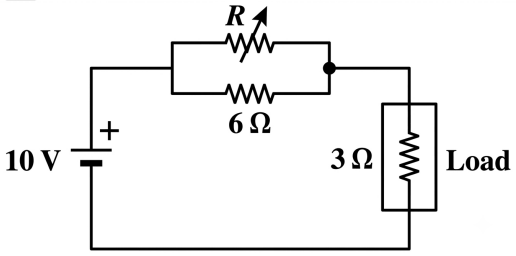


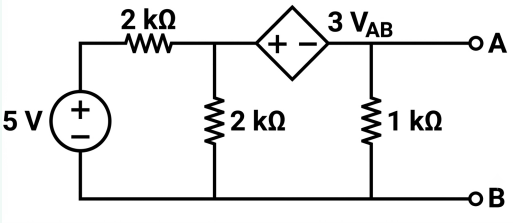
Fig. Q3.11.1

A. Zero **B.** 3 Ω **C.** 6 Ω **D.** Infinity

Common Data Questions

The next 2 questions share the data below.

Statement for Linked Answer Questions Q3.09.1 and Q3.09.2



Q3.09.1 MCQ 2009 2M

For the circuit given above, Thevenin's resistance across the terminals A and B is

A. 0.5 kΩ **B.** 0.2 kΩ **C.** 1 kΩ **D.** 0.11 kΩ

Q3.09.2 MCQ 2009 2M

For the circuit given above, Thevenin's voltage across the terminals A and B is

A. 1.25 V **B.** 0.25 V **C.** 1 V **D.** 0.5 V

Q3.09.3 MCQ 2009 1M

How many 200 W/220 V incandescent lamps connected in series would consume the same total power as a single 100 W/220 V incandescent lamp?

A. Not possible **B.** 4 **C.** 3 **D.** 2

Q3.05.1 MCQ 2005 2M

In the given figure, the Thevenin's equivalent pair (voltage and impedance), as seen at the terminals PQ, is given by

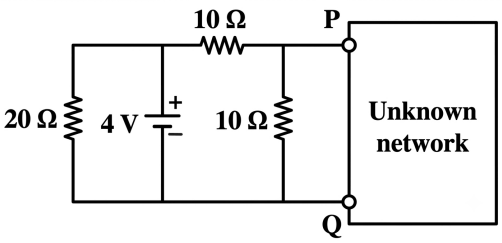


Fig. Q3.05.1

A. 2 V, 5 Ω **B.** 2 V, 7.5 Ω
C. 4 V, 5 Ω **D.** 4 V, 7.5 Ω

Q3.04.1 MCQ 2004 2M

In the following figure, R_a, R_b and R_c are 20 Ω, 10 Ω and 10 Ω, respectively. The resistances R_1, R_2 and R_3 in Ω of an equivalent star-connection are

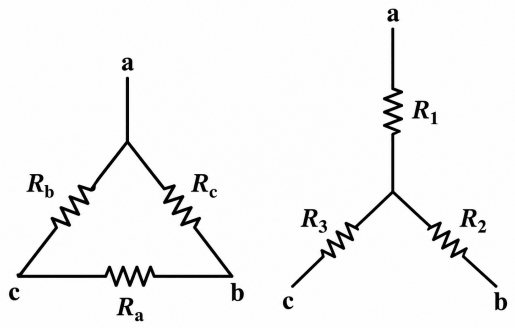


Fig. Q3.04.1

A. 2.5, 5, 5 **B.** 5, 2.5, 5
C. 5, 5, 2.5 **D.** 2.5, 5, 2.5

Q3.03.1 MCQ 2003 2M

In Figure, the potential difference between points P and Q is

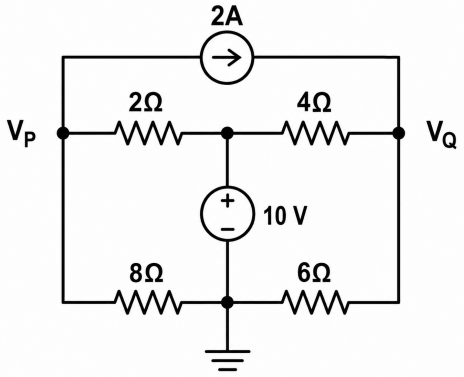


Fig. Q3.03.1

A. 12 V **B.** 10 V **C.** −6 V **D.** 8 V

Q3.02.1 NAT 2002 5M

In the resistor network shown in Figure, all resistor values are 1 Ω. A current of 1 A passes from terminal a to terminal b, as shown in the figure. Calculate the voltage between terminals a and b. [Hint: You may exploit the symmetry of the circuit].

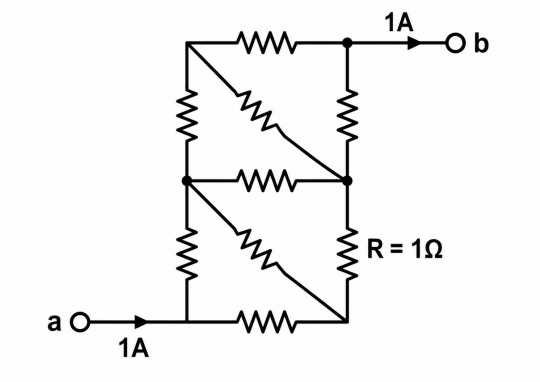


Fig. Q3.02.1

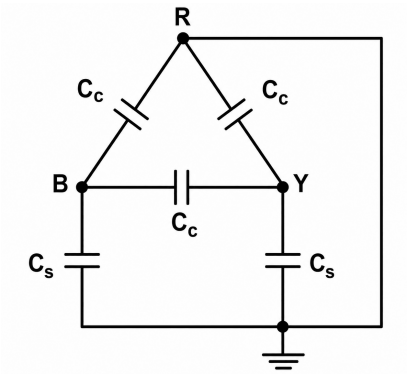


Fig. Q3.99.1

Q3.01.1 MCQ 2001 1M

Two incandescent light bulbs of 40 W and 60 W rating are connected in series across the mains. Then

A. the bulbs together consume 100 W

B. the bulbs together consume 50 W

C. the 60 W bulb glows brighter

D. the 40 W bulb glows brighter

Q3.00.1 MCQ 2000 2M

The circuit shown in figure is equivalent to a load of

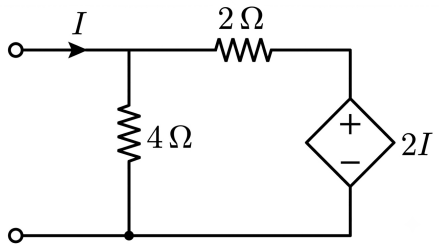


Fig. Q3.00.1

A. $\frac{4}{3}$ ohms **B.** $\frac{8}{3}$ ohms **C.** 4 ohms **D.** 2 ohms

Q3.99.1 MCQ 1999 1M

For the circuit shown in fig. the capacitance measured between terminals B and Y will be

A. $C_c + \left(\frac{C_s}{2}\right)$ **B.** $C_s + \left(\frac{C_c}{2}\right)$

C. $\frac{(C_s + 3C_c)}{2}$ **D.** $3C_c + 2C_s$

Q3.98.1 MCQ 1998 2M

Viewed from the terminals A, B the following circuit shown in figure can be reduced to an equivalent circuit of a single voltage source series with a single resistance with the following parameters:

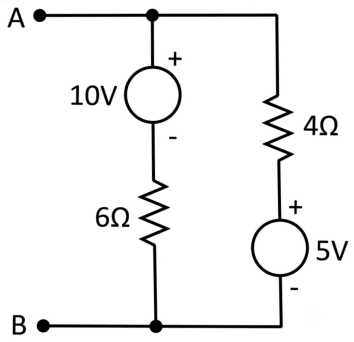


Fig. Q3.98.1

A. 5 volt source in series with 10 Ω resistor

B. 7 volt source in series with 2.4 Ω resistor

C. 15 volt source in series with 2.4 Ω resistor

D. 7 volt source in series with 10 Ω resistor

Q3.97.1 Descriptive 1997 5M

Find the Thevenin equivalent about AB for the circuit shown in figure

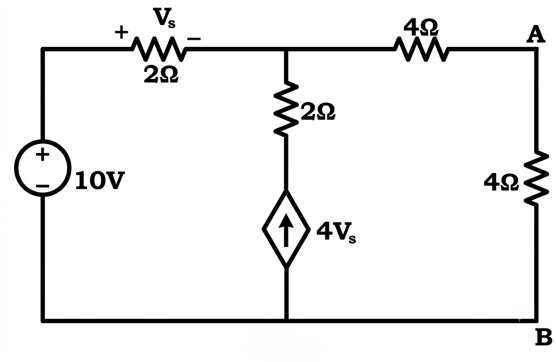


Fig. Q3.97.1

Q3.97.2 Descriptive | 1997 | 2M

For the circuit shown in figure. The Norton equivalent source current value is _____ A and its resistance is _____ Ohms

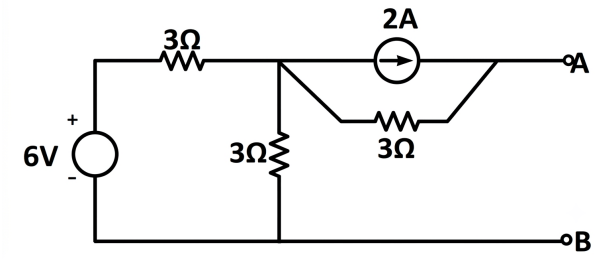


Fig. Q3.97.2

Q3.95.1 Descriptive | 1995 | 5M

For the circuit shown in figure, find the current through the resistance R connected between points a and b by Thevenin's theorem.

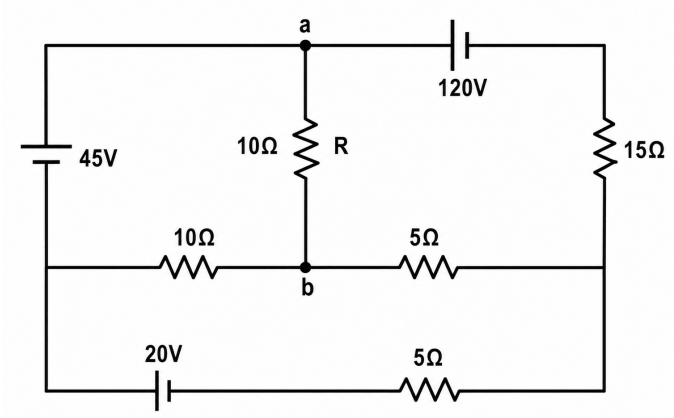


Fig. Q3.95.1

Q3.94.1 MCQ | 1994 | 1M

Superposition principle is not applicable to a network containing time-varying resistors.

A. TRUE

B. FALSE

Q3.92.1 MCQ | 1992 | 1M

All resistances in the figure are 1 Ω each. The value of current 'I' is

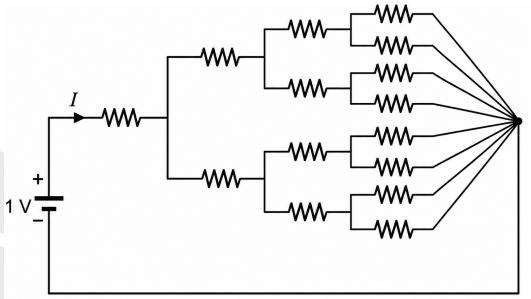


Fig. Q3.92.1

A. $\frac{1}{15}$ A **B.** $\frac{2}{15}$ A **C.** $\frac{4}{15}$ A **D.** $\frac{8}{15}$ A

Q3.91.1 MCQ | 1991 | 1M

The V-I characteristic as seen from the terminal pair (A,B) of the network of figure (a) is shown figure (b), if a variable resistance R_L is connected across the terminal pair (A,B), the maximum power that can be supplied to R_L would be

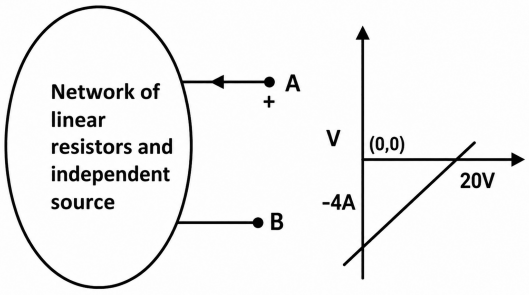


Fig. Q3.91.1

A. 80W

B. 40W

C. 20W

D. Indeterminate unless the actual network is given

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q3.26.1 B	Q3.26.2 C	Q3.26.3 2.81 to 2.91
Q3.21.1 B	Q3.20.1 C	Q3.20.2 13.80 to 14.20
Q3.19.1 0 to 0	Q3.17.1 A	Q3.17.2 3.0 to 3.1
Q3.17.3 2.9 to 3.1	Q3.16.1 D	Q3.16.2 11.25 to 11.50
Q3.15.1 3.2 to 3.5	Q3.14.1 2.9 to 3.1	Q3.14.2 1
Q3.13.1 B	Q3.12.1 A	Q3.12.2 A
Q3.11.1 A	Q3.09.1 B	Q3.09.2 D
Q3.09.3 D	Q3.05.1 A	Q3.04.1 A
Q3.03.1 C	Q3.02.1 *	Q3.01.1 D
Q3.00.1 B	Q3.99.1 C	Q3.98.1 B
Q3.97.1 *	Q3.97.2 2, 4.5	Q3.95.1 *
Q3.94.1 B	Q3.92.1 D	Q3.91.1 C

PrepFusion

UNIT

IV

2nd Order Dynamics and Complex Impedance

Topics

1. Source-Free 2nd Order RLC Circuits
2. Initial Conditions and Sourced RLC Circuits
3. LC Oscillators
4. Equivalent Impedance of AC Circuits
5. AC Transients

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 7	**MCQ:** 2	**MSQ:** 0	**NAT:** 5
1M: 1	**2M:** 6	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2026)–4M(2018)

**See preface for how Avg Weightage and Range are calculated.*

Q4.26.1 NAT 2026 1M

Consider the circuit shown. Assume that the diode D is ideal. The supply voltage $v_s = 325\sin(2\pi 50t)V$, $L = 500\,\mu$H and $R = 10\,\Omega$. The peak diode current (in amperes) is _____ *(Round off to one decimal place)*

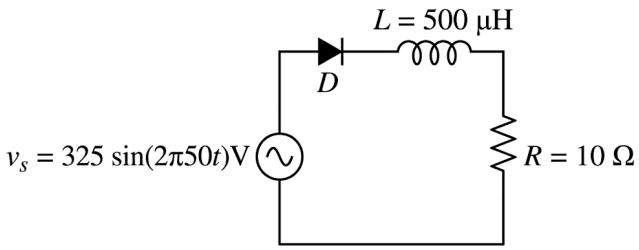


Fig. Q4.26.1

Q4.23.1 NAT 2023 2M

The circuit shown in the figure is initially in the steady state with the switch K in open condition and $\overline{K}$ in closed condition. The switch K is closed and $\overline{K}$ is opened simultaneously at the instant $t = t_1$ where $t_1 > 0$. The minimum value of t_1 in milliseconds such that there is no transient in the voltage across the 100 μF capacitor is (Round off to two decimal places)

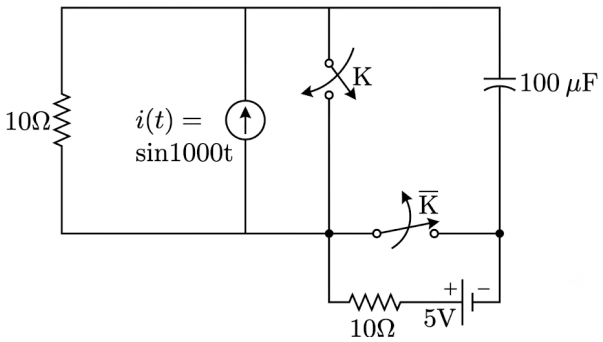


Fig. Q4.23.1

Q4.18.1 MCQ 2018 2M

The equivalent impedance Z_{eq} for the infinite ladder circuit shown in the figure is

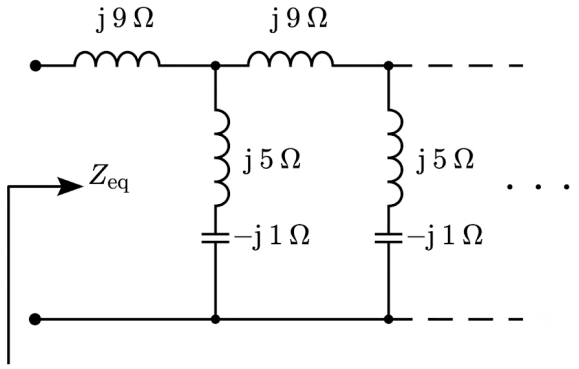


Fig. Q4.18.1

A. $j12\Omega$ **B.** $-j12\Omega$ **C.** $j13\Omega$ **D.** 13Ω

Q4.18.2 NAT 2018 2M

The voltage v(t) across the terminals a and b as shown in the figure, is a sinusoidal voltage having a frequency ω= 100 radian/s. When the inductor current i(t) is in phase with the voltage v(t), the magnitude of the impedance Z (in Ω) seen between the terminals a and b is _____ (up to 2 decimal places).

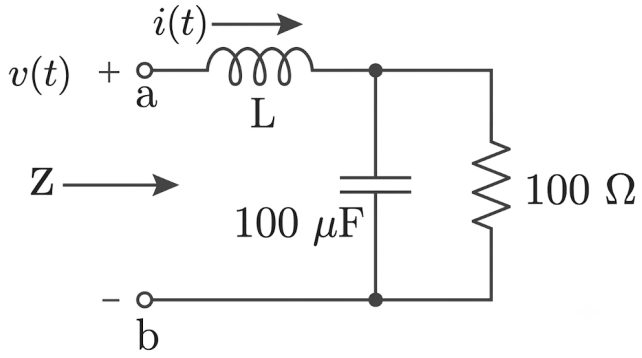


Fig. Q4.18.2

Q4.16.1 NAT 2016 2M

In the circuit shown below, the supply voltage is 10 sin(1000t) volts. The peak value of the steady state current through the 1Ω resistor, in amperes, is _____.

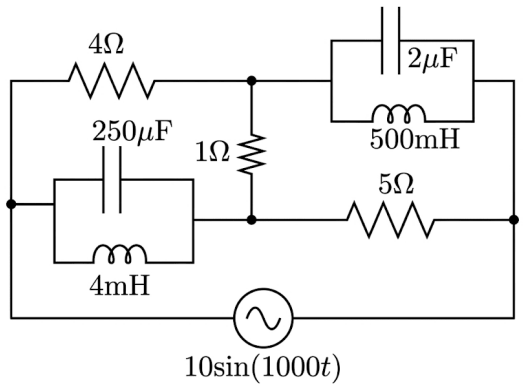


Fig. Q4.16.1

A. 0 ms **B.** 1.78 ms
C. 2.71 ms **D.** 2.91 ms

Answer Key

Q4.26.1 32.3 to 32.5	Q4.23.1 1.57	Q4.18.1 A
Q4.18.2 50.0 to 50.0	Q4.16.1 1.0 to 1.0	Q4.15.1 10
Q4.02.1 B		

Q4.15.1 NAT 2015 2M

The circuit shown in the figure has two sources connected in series. The instantaneous voltage of the AC source (in Volt) is given by v(t) = 12 sin t. If the circuit is in steady state, then the RMS value of the current (in Ampere) flowing in the circuit is _____.

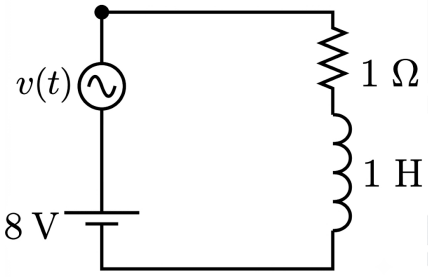


Fig. Q4.15.1

Q4.02.1 MCQ 2002 2M

Consider the circuit shown in the Figure. If the frequency of the source is 50 Hz, then a value of t_0 which results in a transient free response is

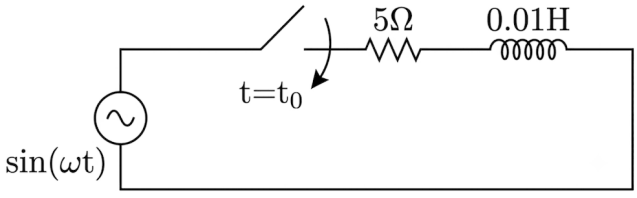


Fig. Q4.02.1

UNIT

V

AC Circuit Analysis

Topics

1. Basics of Phasor Diagrams
2. Phasor Diagram for Series RL, RC and RLC Circuits
3. Phasor Diagram for Parallel RL, RC and RLC Circuits
4. Locus Diagram of Current, Admittance and Impedance
5. Application of Network Theorems in AC Circuits
6. Maximum Power Transfer in AC Circuits

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 41	**MCQ:** 27	**MSQ:** 0	**NAT:** 14
1M: 17	**2M:** 21	**Descriptive:** 3	**Avg Weightage** ~2M	**Range:** 1M(2020)–6M(2011)

**See preface for how Avg Weightage and Range are calculated.*

Q5.25.1 NAT 2025 2M

In an experiment to measure the active power drawn by a single-phase RL Load connected to an AC source through a 2Ω resistor, three voltmeters are connected as shown in the figure below. The voltmeter readings are as follows: $V_{Source} = 200V, V_R = 9V, V_{Load} = 199V$. Assuming perfect resistors and ideal voltmeters, the Load-active power measured in this experiment, in W, is _____. *(round off to one decimal place).*

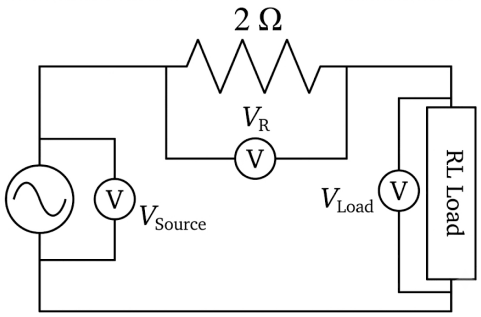


Fig. Q5.25.1

Q5.24.1 NAT 2024 2M

For the circuit shown in the figure, the source frequency is 5000 rad/sec. The mutual inductance between the magnetically coupled inductors is 5 mH with their self inductances being 125 mH and 1 mH. The Thevenin's impedance, Z_{th} between the terminals P and Q in Ω is _____ (rounded off to 2 decimal places).

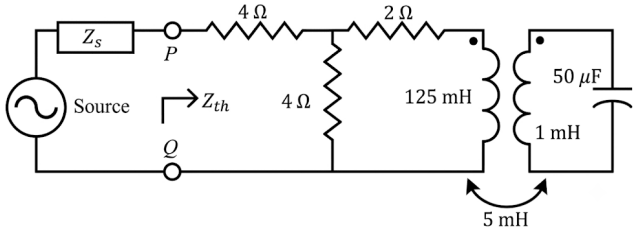


Fig. Q5.24.1

Q5.23.1 NAT 2023 2M

For the circuit shown, if $i = \sin 1000t$, the instantaneous value of the Thevenin's equivalent voltage (in Volts) across the terminals a-b at time t = 5 ms is _____ (Round off to 2 decimal places).

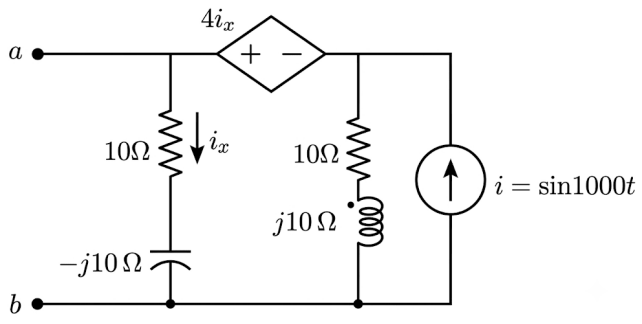


Fig. Q5.23.1

Q5.21.1 NAT 2021 1M

In the given circuit, the value of capacitor C that makes current I=0 is _____ μF.

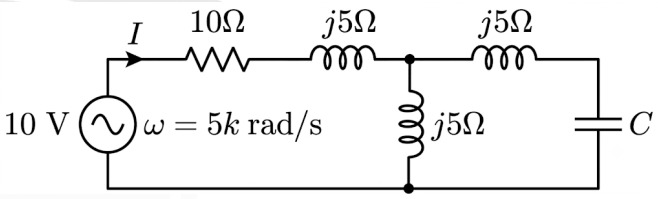


Fig. Q5.21.1

Q5.21.2 NAT 2021 1M

A signal generator having a source resistance of 50Ω is set to generate a 1kHz sinewave. Open circuit terminal voltage is 10V peak-to-peak. Connecting a capacitor across the terminals reduces the voltage to 8V peak-to-peak. The value of this capacitor is _____ μF. (Round off to 2 decimal places.)

Q5.21.3 NAT 2021 1M

In the given circuit, for maximum power to be delivered to R_L, its value should be _____ Ω. (Round off to 2 decimal places.)

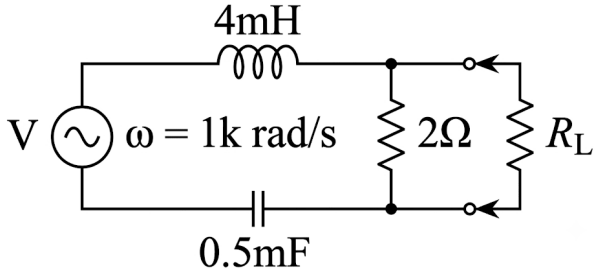


Fig. Q5.21.3

Q5.20.1 NAT 2020 1M

Currents through ammeters A2 and A3 in the figure are $1\angle 10°$ and $1\angle 70°$, respectively. The reading of the ammeter A1 (rounded off to 3 decimal places) is _____ A.

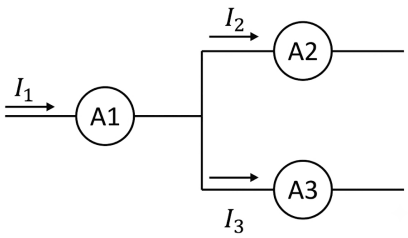


Fig. Q5.20.1

Q5.18.1 MCQ 2018 2M

A DC voltage source is connected to a series L-C circuit by turning on switch S at time t=0 as shown in the figure. Assume i(0)=0, v(0)=0. Which one of the following circular loci represents the plot of i(t) versus v(t)?

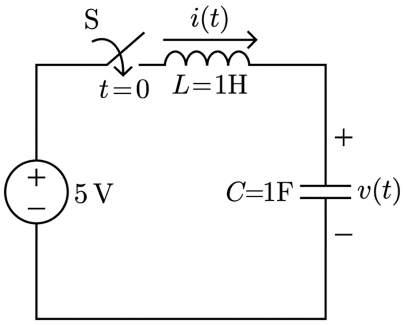
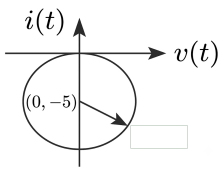
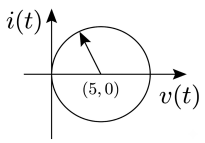
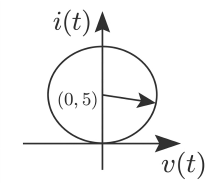
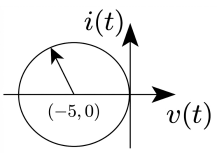


Fig. Q5.18.1

A.


B.


C.


D.


Q5.17.2 MCQ 2017 2M

In the circuit show below, the value of capacitor C required for maximum power to be transferred to the load is

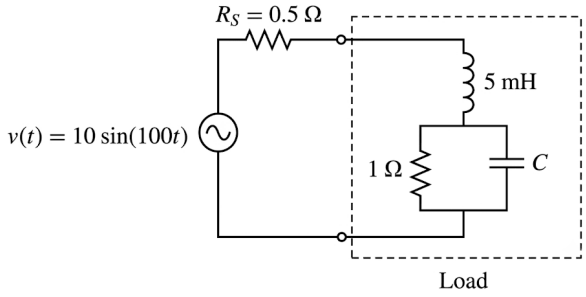


Fig. Q5.17.2

A. 1 nF **B.** 1 μF **C.** 1 mF **D.** 10 mF

Q5.16.1 MCQ 2016 1M

A resistance and a coil are connected in series and supplied from a single phase 100V, 50Hz ac source as shown in the figure below. The rms values of plausible voltages across the resistance (V_R) and coil (V_C) respectively, in volts, are

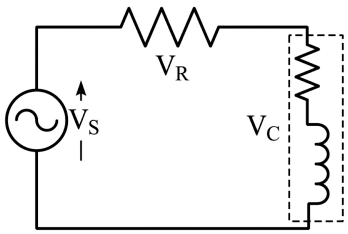


Fig. Q5.16.1

A. 65,35 **B.** 50,50
C. 60,90 **D.** 60,80

Q5.15.1 MCQ 2015 1M

An inductor is connected in parallel with a capacitor as shown in the figure.

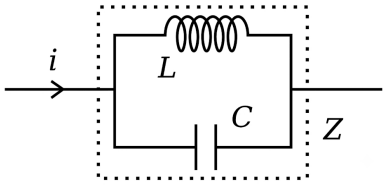
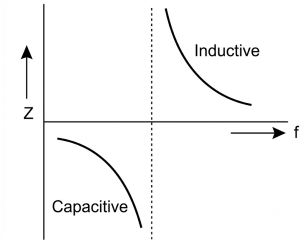
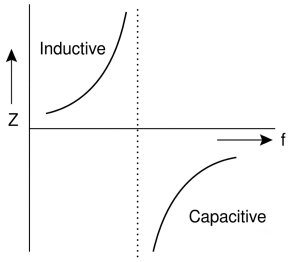
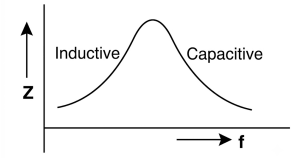


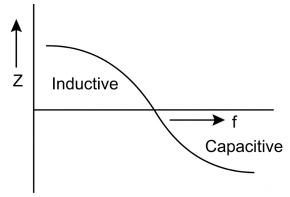
Fig. Q5.15.1

As the frequency of current i is increased, the impedance (Z) of the network varies as

A.


B.


C.


D.


Q5.14.1 MCQ 2014 2M

The voltage across the capacitor, as shown in the figure, is expressed as

$$v_c(t) = A_1 \sin(\omega_1 t - \theta_1) + A_2 \sin(\omega_2 t - \theta_2)$$

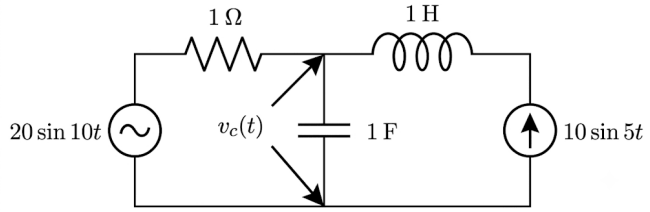


Fig. Q5.14.1

The values of A_1 and A_2 respectively, are

A. 2.0 and 1.98
B. 2.0 and 4.20
C. 2.5 and 3.50
D. 5.0 and 6.40

Q5.14.2 MCQ 2014 2M

A series RLC circuit is observed at two frequencies. At $\omega_1 = 1$ krad/s, we note that source voltage $V_1 = 100\angle 0°$ V results in a current $I_1 = 0.03\angle 31°$ A. At $\omega_2 = 2$ krad/s, the source voltage $V_2 = 100\angle 0°$ V results in a current $I_2 = 2\angle 0°$ A. The closest values for R,L,C out of the following options are

A. $R = 50\Omega, L = 25\,mH, C = 10\mu F$

B. $R = 50\Omega, L = 10\,mH, C = 25\mu F$

C. $R = 50\Omega, L = 50\,mH, C = 5\mu F$

D. $R = 50\Omega, L = 5\,mH, C = 50\mu F$

Q5.13.1 MCQ 2013 2M

In the circuit shown below, if the source voltage $100\angle 53.13°$ V then the Thevenin's equivalent voltage in Volts as seen by the load resistance R_L is

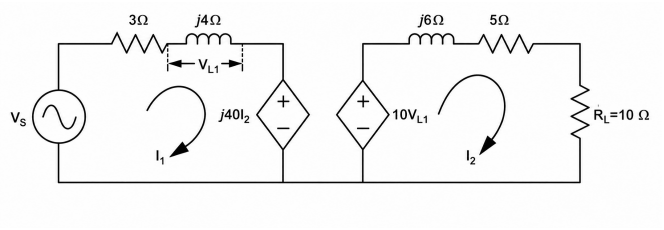


Fig. Q5.13.1

A. $100\angle 90°$
B. $800\angle 0°$
C. $800\angle 90°$
D. $100\angle 60°$

Q5.12.1 MCQ 2012 1M

In the circuit shown below, the current through the inductor is

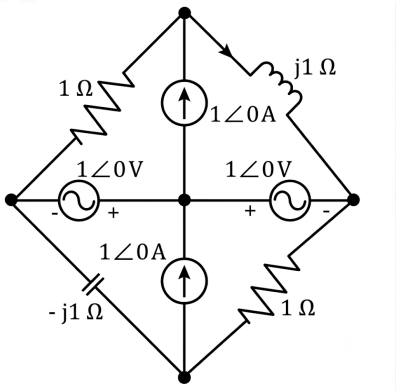


Fig. Q5.12.1

A. $\frac{2}{1+j}$ A
B. $\frac{-1}{1+j}$ A
C. $\frac{1}{1+j}$ A
D. 0 A

Q5.11.1 MCQ 2011 1M

The r.m.s value of the current i(t) in the circuit shown below is

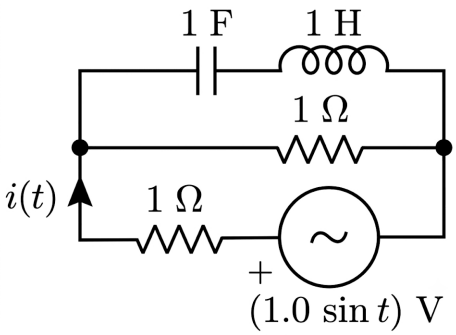


Fig. Q5.11.1

A. $\frac{1}{2}A$ **B.** $\frac{1}{\sqrt{2}}A$ **C.** 1A **D.** $\sqrt{2}A$

Q5.11.2 MCQ 2011 1M

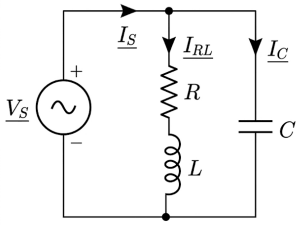
The voltage applied to a circuit is $100\sqrt{2}\cos(100\pi t)$ volts and the circuit draws a current of $10\sqrt{2}\sin(100\pi t + \pi/4)$ amperes. Taking the voltage as the reference phasor, the phasor representation of the current in amperes is

A. $10\sqrt{2}\angle -\pi/4$ **B.** $10\angle -\pi/4$
C. $10\angle +\pi/4$ **D.** $10\sqrt{2}\angle +\pi/4$

Common Data Questions

The next 2 questions share the data below.

Common Data for Questions 5.11.3 and 5.11.4:
An RLC circuit with relevant data is given below.



$\underline{V_S} = 1\angle 0$ V
$\underline{I_S} = \sqrt{2}\angle \pi/4$ A
$\underline{I_{RL}} = \sqrt{2}\angle -\pi/4$ A

Q5.11.3 MCQ 2011 2M

The power dissipated in the resistor R is

A. 0.5W **B.** 1W **C.** $\sqrt{2}W$ **D.** 2W

Q5.11.4 MCQ 2011 2M

The current $\underline{I_C}$ in the figure above is

A. -j2 A **B.** $-j\frac{1}{\sqrt{2}}$ A
C. $+j\frac{1}{\sqrt{2}}$ A **D.** +j2 A

Q5.08.1 MCQ 2008 1M

The Thevenin's equivalent of a circuit operating at ω= 5 rad/s, has $V_{oc} = 3.71\angle -15.9°$ V and $Z_o = 2.38 - j0.667\Omega$. At this frequency, the minimal realization of the Thevenin's impedance will have a

A. resistor and a capacitor and an inductor

B. resistor and a capacitor

C. resistor and an inductor

D. capacitor and an inductor

Q5.07.1 MCQ 2007 2M

The R-L-C series circuit shown is supplied from a variable frequency voltage source. The admittance-locus of the R-L-C network at terminals ABfor increasing frequency ω is

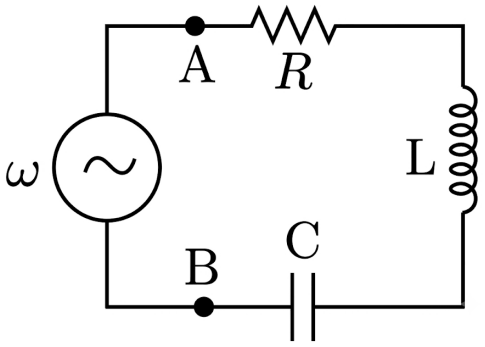
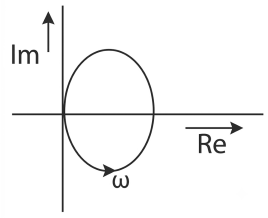
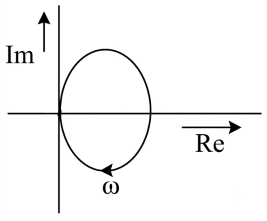
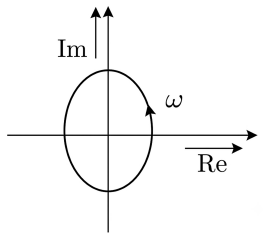
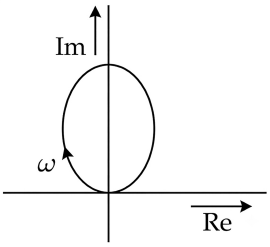


Fig. Q5.07.1

A.


B.


C.


D.


Q5.07.2 MCQ 2007 2M

In the figure given below all phasors are in reference to the potential at point "O".The locus of voltage phasor V_{YX} as R is varied from zero to infinity is shown by

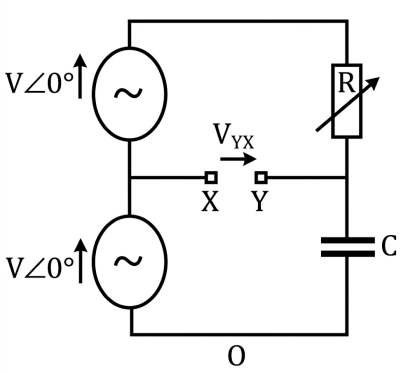
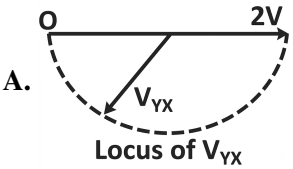
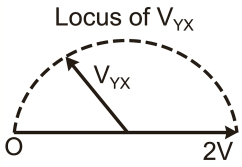
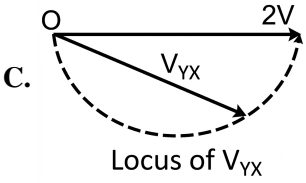
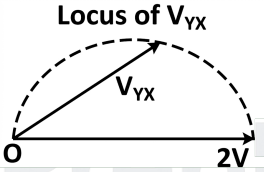


Fig. Q5.07.2

A. 

B. 

C. 

D. 

Q5.06.1 MCQ 2006 1M

In the figure the current source is $1\angle 0°$ A, R = 1Ω. the impedances are $Z_C = -j\Omega$ and $Z_L = 2j\Omega$. The Thevenin equivalent looking into the circuit across X-Y is

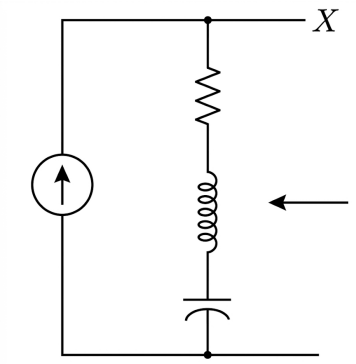


Fig. Q5.06.1

A. $\sqrt{2}\angle 0°V$,(1+2j)Ω **B.** $2\angle 45°V$,(1-2j)Ω
C. $2\angle 45°V$,(1+j)Ω **D.** $\sqrt{2}\angle 45°V$,(1+j)Ω

Q5.06.2 MCQ 2006 2M

The circuit shown in the figure is energized by a sinusoidal voltage source V_1 at a frequency which causes resonance with a current of I.

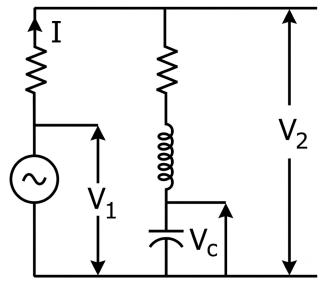
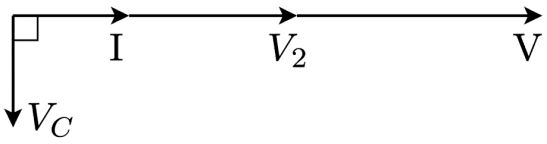
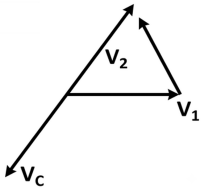
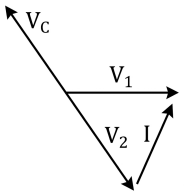
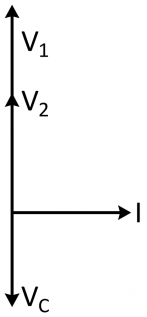


Fig. Q5.06.2

A. 

B. 

C. 

D. 

Q5.05.1 MCQ 2005 2M

The RL circuit of Figure is fed from a constant magnitude, variable frequency sinusoidal voltage source V_{IN}. At 100 Hz, the R and L elements each have a voltage drop U_{RMS}. If the frequency of the source is changed to 50 Hz, the new voltage drop across R is :

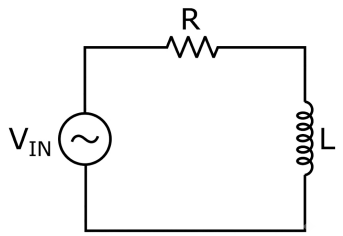


Fig. Q5.05.1

A. $\sqrt{\frac{5}{8}}U_{RMS}$ **B.** $\sqrt{\frac{2}{3}}U_{RMS}$

C. $\sqrt{\frac{8}{5}}U_{RMS}$ **D.** $\sqrt{\frac{3}{2}}U_{RMS}$

Q5.04.1 MCQ 2004 2M

In figure, the admittance values of the elements in Siemens are

$Y_R = 0.5 + j0, Y_L = 0 - j1.5, Y_C = 0 + j0.3$ respectively

The value of I as a phasor when the voltage E across the elements is $10\angle 0°V$ is

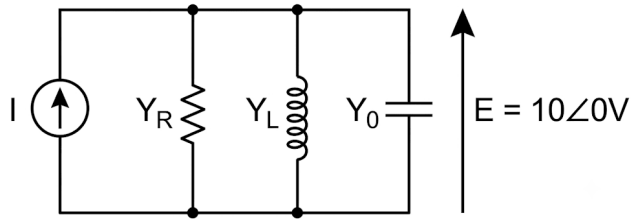


Fig. Q5.04.1

A. 1.5+j0.5 **B.** 5-j18

C. 0.5+j1.8 **D.** 5-j12

Q5.03.1 MCQ 2003 1M

A segment of a circuit is shown in Figure. $V_R = 5V, V_C = 4\sin 2t$. The voltage V_L is given by

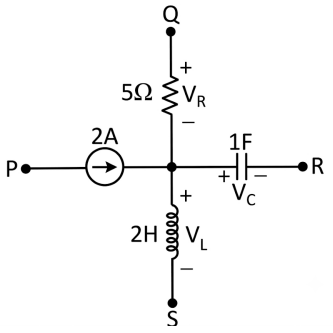


Fig. Q5.03.1

A. 3-8 cos 2t **B.** 32 sin 2t

C. 16 sin 2t **D.** 16 cos 2t

Q5.03.2 MCQ 2003 1M

In the Figure,

$$Z_1 = 10\angle -60°, Z_2 = 10\angle 60°. Z_3 = 50\angle 53.13°$$

The thevenin impedance seen from X-Y is

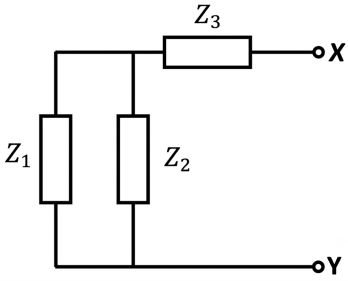


Fig. Q5.03.2

A. $56.6\angle 45°$ **B.** $60\angle 30°$

C. $70\angle 30°$ **D.** $34.4\angle 65°$

Q5.02.1 NAT 2002 5M

An electrical network is fed by two ac sources, as shown in Figure. Given that $Z_1 = (1 - j)\Omega, Z_2 = (1 + j)\Omega$ and $Z_L = (1 - j0)\Omega$. Obtain the Thevenin equivalent circuit (Thevenin voltage and impedance) across terminals x and y, and determine current I_L through the load Z_L.

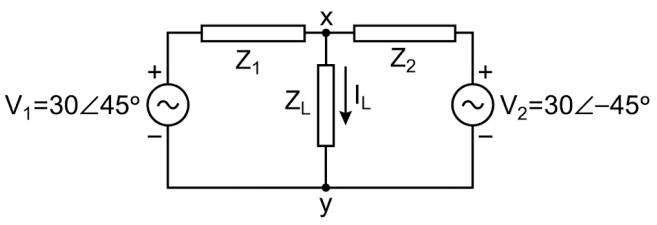


Fig. Q5.02.1

Q5.00.1 MCQ 2000 2M

The impedance seen by the source in the circuit in figure, is given by

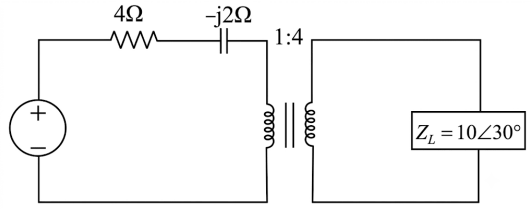


Fig. Q5.00.1

A. (0.54+j0.313) ohms

B. (4-j2) ohms

C. (4.54-j1.69) ohms

D. (4+j2) ohms

Q5.00.2 NAT 2000 5M

Predict the current I in figure in response to a voltage of $20\angle 0°V$. The impedance values are given in ohms. Use the thevenin's theorem.

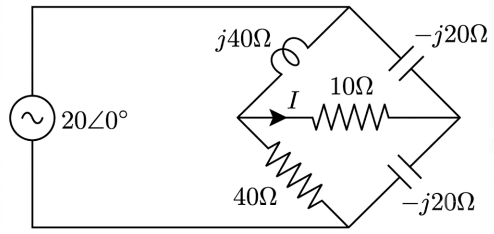


Fig. Q5.00.2

Q5.99.1 MCQ 1999 2M

Currents I_1,I_2 and I_3 meet at a junction (node) in a circuit. All currents are marked as entering the node. If $I_1 = -6\sin(\omega t)$ mA and $I_2 = 8\cos(\omega t)$ mA , then I_3 will be

A. $10\cos(\omega t + 36.87)$ mA

B. $14\cos(\omega t + 36.87)$ mA

C. $-14\sin(\omega t + 36.87)$ mA

D. $-10\cos(\omega t + 36.87)$ mA

Q5.99.2 MCQ 1999 2M

A fixed capacitor of reactance $-j0.02\,k\Omega$ is connected in parallel across a series combination of a fixed inductor of reactance $j0.01\,k\Omega$ and a variable resistance R. As R is varied from zero to infifnity, the locus diagram of the admittance of this L-C-R circuit will be

A. A semi-circle of diameter j100 and centre at zero

B. A semi-circle of diameter j50 and centre at zero

C. A straight line inclined at an angle

D. A straight line parallel to the x-axis

Q5.99.3 NAT 1999 2M

A constant voltage frequency sinusoidal voltage source of magnitude V_s is connected to a series circuit made of a resistance 15 Ω, a coil of winding resistance R and inductance L and a 50 μF capacitor. The voltage across the 15 Ω resistor is 30V, across the coil is 50V, across the capacitor is 40V. The voltage across the combination of the 15 Ω resistor and the coil together is 72.11V. Determine the values of the inductance L, winding resistance R and the source voltage V.

Q5.98.1 MCQ 1998 1M

A sinusoidal source of voltage V and frequency f is connected to a series circuit of variable resistance, R and a fixed reactance, X. The locus of the tip of the current-phasor I, as R is varied from 0 to ∞ is :

A. A semicircle with a diameter of $\frac{V}{X}$

B. A straight line with a slope of $\frac{R}{X}$

C. An ellipse with $\frac{V}{R}$ as major axis

D. A circle of radius $\frac{R}{X}$ and origin at $(0, \frac{V}{2})$

Q5.95.1 NAT 1995 5M

In the network system shown in figure, find the current through Z_3 using nodal method. The values of voltages are given in volts and the impedances are given in ohms.

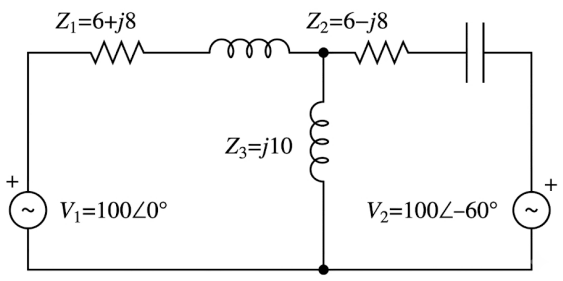


Fig. Q5.95.1

Q5.95.2 NAT 1995 1M

In the circuit shown in figure, ammeter A_2 reads 12 A and A_3 reads 9 A. A_1 will read _____ A.

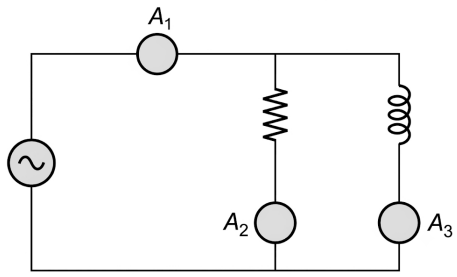


Fig. Q5.95.2

Q5.94.1 NAT 1994 1M

In the given circuit, the voltage $\bar{V}_L$, has a phase angle of _____ with respect to $\bar{V}_S$.

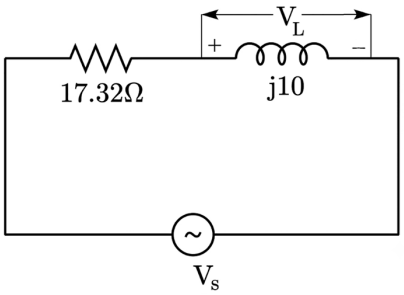


Fig. Q5.94.1

Q5.93.1 MCQ 1993 1M

In the following circuit, i(t) under steady state is

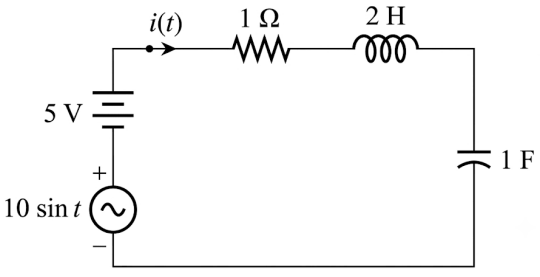


Fig. Q5.93.1

A. zero
B. 5
C. 7.07 sin t
D. 7.07 sin$(t - 45°)$

Q5.91.1 NAT 1991 2M

A one port active network has an input admittance Y, the magnitude of which is shown in figure as a function of frequency. The circuit is resistive or capacitive in different frequency ranges. Complete the following table:

Frequency Range	Type of Impedance	Value (Ω/H/F)
10000 rad/sec < ω	—	—
10 rad/sec < ω < 1000 rad/sec	—	—

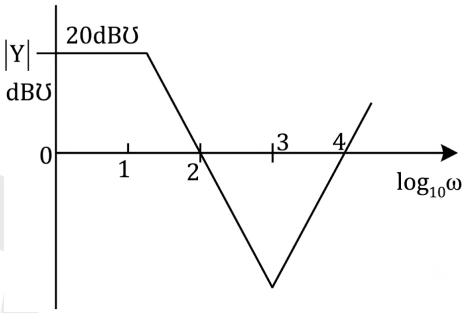


Fig. Q5.91.1

Q5.91.2 MCQ 1991 2M

In figure shown, A_1, A_2 and A_3 are ideal ammeters. If A_1 and A_3 read 5 and 13 A respectively, reading of A_2 will be

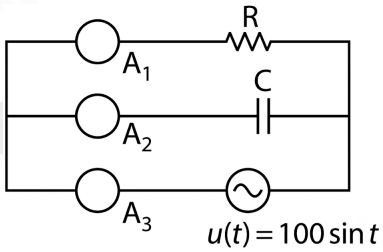


Fig. Q5.91.2

A. 8A

B. 12A

C. 18A

D. indeterminate unless the actual values of R, C and ω are specified

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.
MTA Marks To All awarded — see detailed solutions.

Q5.25.1 78.0 to 81.0	Q5.24.1 5.32 to 5.34	Q5.23.1 -11.98
Q5.21.1 20.00 to 20.00	Q5.21.2 2.30 to 2.50	Q5.21.3 1.40 to 1.42
Q5.20.1 1.700 to 1.750	Q5.18.1 B	Q5.17.2 D
Q5.16.1 MTA	Q5.15.1 B	Q5.14.1 A
Q5.14.2 *	Q5.13.1 C	Q5.12.1 C
Q5.11.1 B	Q5.11.2 B	Q5.11.3 B
Q5.11.4 D	Q5.08.1 B	Q5.07.1 B
Q5.07.2 A	Q5.06.1 D	Q5.06.2 A
Q5.05.1 C	Q5.04.1 D	Q5.03.1 B
Q5.03.2 A	Q5.02.1 *	Q5.00.1 C
Q5.00.2 *	Q5.99.1 D	Q5.99.2 A
Q5.99.3 *	Q5.98.1 A	Q5.95.1 *
Q5.95.2 15	Q5.94.1 60	Q5.93.1 D
Q5.91.1 *	Q5.91.2 B	

UNIT

VI

Resonance

Topics

1. Series RLC Resonance Circuit
2. Quality Factor and Stored Energy in Series Resonance
3. Parallel RLC Resonance Circuit

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 18	**MCQ:** 14	**MSQ:** 0	**NAT:** 4
1M: 12	**2M:** 6	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2025)–2M(2022)

**See preface for how Avg Weightage and Range are calculated.*

Q6.25.1 MCQ 2025 1M

A coil of inductance 100 mH and series resistance 100Ω is connected across a 20 V, 1 kHz supply. The quality factor of the coil at 1 kHz is [Buffer Question]

A. π **B.** 2π **C.** 10^{-3} **D.** 1

Q6.22.1 NAT 2022 1M

An inductor having a Q-factor of 60 is connected in series with a capacitor having a Q- factor of 240. The overall Q-factor of the circuit is _____. *round off to nearest integer*

Q6.22.2 NAT 2022 1M

The network shown below has a resonant frequency of 150 kHz and a bandwidth of 600 Hz. The Q-factor of the network is _____. *round off to nearest integer*

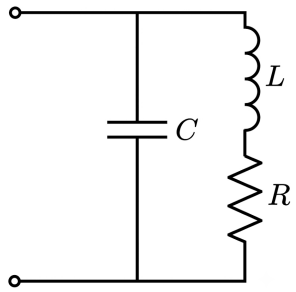


Fig. Q6.22.2

Q6.16.1 NAT 2016 2M

The circuit below is excited by a sinusoidal source. The value of R, in Ω, for which the admittance of the circuit becomes a pure conductance at all frequencies is _____.

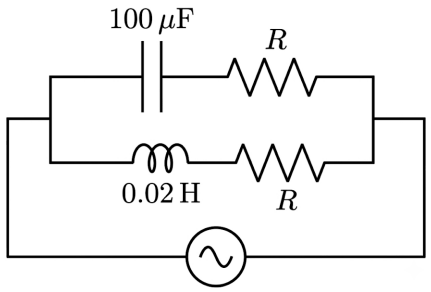


Fig. Q6.16.1

Q6.13.1 MCQ 2013 2M

Two magnetically uncoupled inductive coils have Q factors q_1 and q_2 at the chosen operating frequency. Their respective resistances are R_1 and R_2. When connected in series, their effective Q factor at the same operating frequency is

A. $q_1R_1 + q_2R_2$ **B.** $q_1/R_1 + q_2/R_2$
C. $(q_1R_1 + q_2R_2)/(R_1 + R_2)$ **D.** $q_1R_2 + q_2R_1$

Q6.08.1 MCQ 2008 1M

The resonant frequency for the given circuit will be

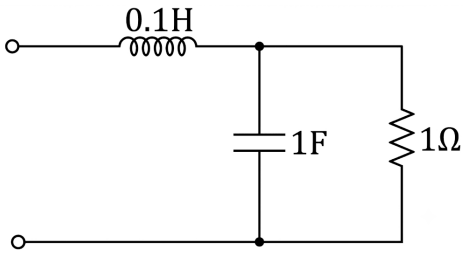


Fig. Q6.08.1

A. 1 rad/s **B.** 2 rad/s
C. 3 rad/s **D.** 4 rad/s

Q6.04.1 MCQ 2004 1M

In the figure the value of Z in Figure, which is most appropriate to cause parallel resonance at 500 Hz, is

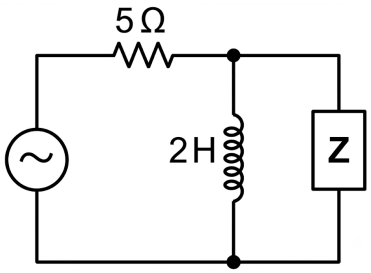


Fig. Q6.04.1

A. 125.00 mH **B.** 304.20 μF
C. 2.0 μF **D.** 0.05 μF

Q6.03.1 MCQ 2003 2M

In the circuit of Figure, the magnitudes of V_L and V_C are twice that of V_R. The inductance of the coil is

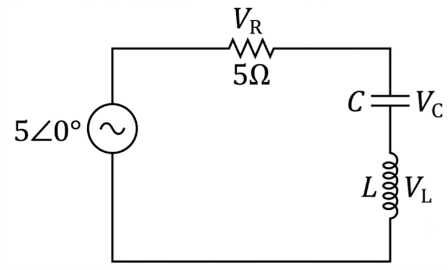


Fig. Q6.03.1

A. 2.14 mH **B.** 5.30 mH
C. 31.84 mH **D.** 1.32 mH

Q6.02.1 MCQ 2002 2M

A series R-L-C circuit has $R = 50\Omega$, $L = 100\,\mu H$ and $C = 1\,\mu F$. The lower half power frequency of the circuit is

A. 30.55 kHz **B.** 3.055 kHz
C. 51.92 kHz **D.** 1.92 kHz

Q6.02.2 MCQ 2002 2M

In the circuit shown in Figure, what value of C will cause a unity power factor at the ac source?

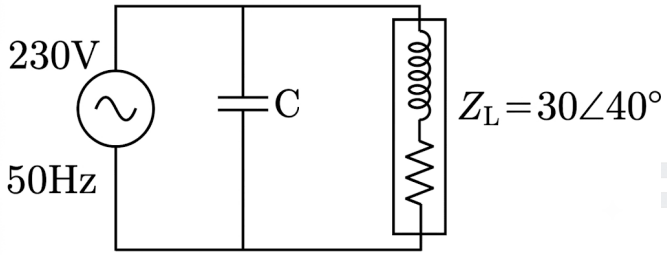


Fig. Q6.02.2

A. 68.1 μF **B.** 165 μF
C. 0.681 μF **D.** 6.81 μF

Q6.01.1 MCQ 2001 1M

In a series RLC circuit at resonance, the magnitude of the voltage developed across the capacitor

A. is always zero

B. can never be greater than the input voltage

C. can be greater than the input voltage, however it is 90° out of phase with the input voltage

D. can be greater than the input voltage, and is in phase with the input voltage

Q6.99.1 MCQ 1999 1M

A series R-L-C circuit when excited by a 10V sinusoidal voltage source of variable frequency, exhibits resonance at 100Hz and has a 3 dB bandwidth of 5 Hz. The voltage across the inductor L at resonance is :

A. 10 V **B.** $10\sqrt{2}$ V
C. $\frac{10}{\sqrt{2}}$ V **D.** 200 V

Q6.99.2 MCQ 1999 1M

The current in the circuit shown in figure is :

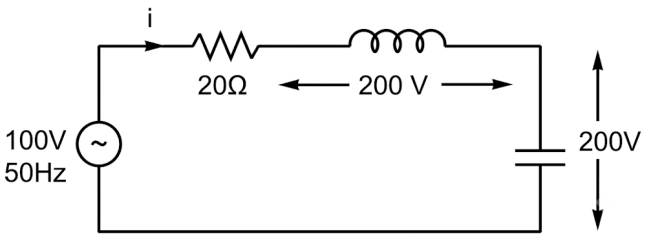


Fig. Q6.99.2

A. 5A **B.** 10A **C.** 15A **D.** 25A

Q6.98.1 MCQ 1998 1M

A circuit with a resistor, inductor and capacitor in series is resonant of f_0 Hz. If all component values are now doubled, the new resonant frequency is :

A. $2f_0$ **B.** still f_0 **C.** $\frac{f_0}{4}$ **D.** $\frac{f_0}{2}$

Q6.96.1 MCQ 1996 2M

A coil (which can be modeled as a series RL circuit) has been designed for high - Q performance at a rated voltage and a specified frequency. If the frequency of operation is doubled, and the coil is operated at the same rated voltage, then the Q-factor and the active power P consumed by the coil will be affected as follows

A. P is doubled, Q is halved

B. P is halved, Q is doubled

C. P remains constant, Q is doubled

D. P is decreased 4 times, Q is doubled

Q6.95.1 NAT 1995 1M

A series R-L-C circuit has the following parameter values. $R = 10\,\Omega$, L=0.01 H, C= 100 mF. The Q factor of the circuit at resonance is _____

Q6.94.1 MCQ 1994 1M

At resonance, the given parallel circuit constituted by an iron-coil and a capacitor behaves like

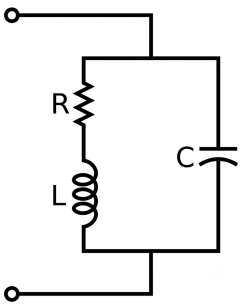


Fig. Q6.94.1

A. An open-circuit

B. A short-circuit

C. A pure resistor of value R

D. A pure resistor of value much higher than R

Q6.93.1 MCQ 1993 1M

The following circuit (figure) resonates at

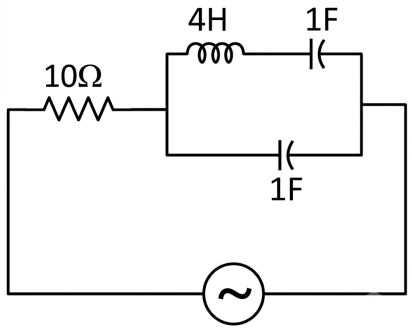


Fig. Q6.93.1

A. all frequencies
B. 0.5 rad/sec
C. 5 rad/sec
D. 1 rad/sec

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q6.25.1 B	Q6.22.1 48 to 48	Q6.22.2 250 to 250
Q6.16.1 14.0 to 14.2	Q6.13.1 C	Q6.08.1 C
Q6.04.1 D	Q6.03.1 *	Q6.02.1 B
Q6.02.2 A	Q6.01.1 C	Q6.99.1 D
Q6.99.2 A	Q6.98.1 D	Q6.96.1 D
Q6.95.1 0.03162	Q6.94.1 D	Q6.93.1 B

UNIT

VII

Complex Power

Topics

1. Introduction to Power in AC Circuits
2. Active, Reactive and Apparent Power in Complex Networks

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 20	**MCQ:** 14	**MSQ:** 0	**NAT:** 6
1M: 10	**2M:** 10	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2025)–7M(2012)

**See preface for how Avg Weightage and Range are calculated.*

Q7.26.1 MCQ 2026 1M

A single-phase voltage source $v_s(t) = 325\sin(2\pi 50t)V$, delivers a current, $i = 12\sin(2\pi 50t) + 9\sin(2\pi 150t)A$ to a load. The load power factor, correct up to two decimal places, is:

A. 1.00 **B.** 0.80 **C.** 0.65 **D.** 0.57

Q7.26.2 MCQ 2026 2M

An electrical component has voltage drop $v = V_m\sin(\omega t)$, when the current through it is $i = I_m\sin(\omega t - \theta)$. What is the average power dissipated over a half cycle corresponding to ω ?

A. 0 **B.** $V_m I_m cos\theta$

C. $\frac{V_m I_m}{2} cos\theta$ **D.** $\frac{V_m I_m}{4} cos\theta$

Q7.25.1 NAT 2025 1M

A device (single-phase, 50 Hz, AC) yields the following measurements at its terminals: voltage = $141.4\angle -10°V$, current = $0.7\angle -55°A$.The power factor of the device is _____. *(round off to three decimal places).*

Q7.19.1 NAT 2019 2M

The voltage across and the current through a load are expressed as follows

$$v(t) = -170\sin(377t - \frac{\pi}{6})V$$

$$i(t) = 8\cos(377t + \frac{\pi}{6})A$$

The average power in watts (round off to one decimal place) consumed by the load is _____.

Q7.18.1 MCQ 2018 1M

In the figure, the voltages are $v_1(t) = 100\cos(\omega t)$, $v_2(t) = 100\cos(\omega t + \pi/18)$ and $v_3(t) = 100\cos(\omega t + \pi/36)$. The circuit is in sinusoidal steady state, and $R << \omega L$. P_1,P_2 and P_3 are the average power outputs. Which one of the following statements is true?

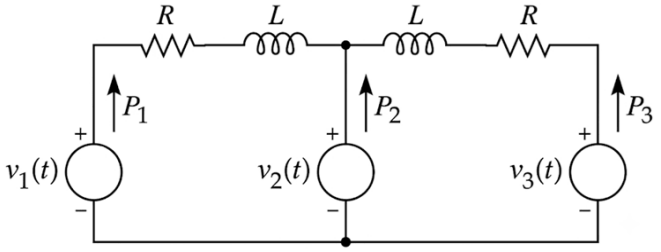


Fig. Q7.18.1

A. $P_1 = P_2 = P_3 = 0$ **B.** $P_1 < 0, P_2 > 0, P_3 > 0$

C. $P_1 < 0, P_2 > 0, P_3 < 0$ **D.** $P_1 > 0, P_2 < 0, P_3 > 0$

Q7.18.2 NAT 2018 2M

The voltage across the circuit in the figure, and the current through it, are given by the following expressions :

$$v(t) = 5 - 10\cos(\omega t + 60°)V$$

$$i(t) = 5 + X\cos(\omega t)A$$

where $\omega = 100\pi$radian/s. If the average power delivered to the circuit is zero, then the value of X (in Ampere) is _____ (up to 2 decimal places).

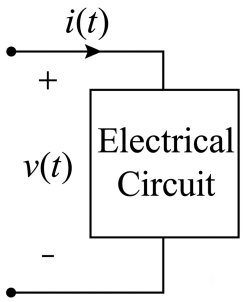


Fig. Q7.18.2

Q7.17.1 MCQ 2017 1M

A source is supplying a load through a 2-phase, 3-wire transmission system as shown in figure below. The instantaneous voltage and current in phase - a are $v_{an} = 220\sin(100\pi t)V$ and $i_a = 10\sin(100\pi t)A$, respectively. Similarly for phase-b, The instantaneous voltage and current are $v_{bn} = 220\cos(100\pi t)V$ and $i_b = 10\cos(100\pi t)A$, respectively.

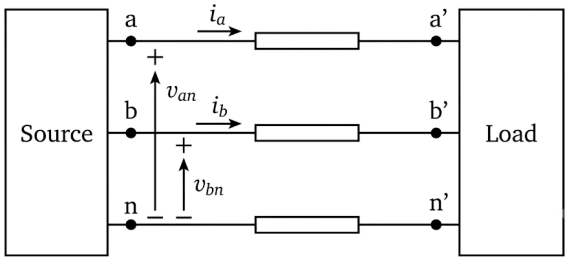


Fig. Q7.17.1

The total instantaneous power flowing from the source to the load is

A. 2200W **B.** $2200\sin^2(100\pi t)$ W

C. 4400W **D.** $2200\sin(100\pi t)\cos(100\pi t)$ W

Q7.16.1 NAT 2016 1M

The voltage(V) and current (A) across a load are as follows.

$$v(t) = 100\sin(\omega t),$$

$$i(t) = 10\sin(\omega t - 60^\circ) + 2\sin(3\omega t) + 5\sin(5\omega t)$$

The average power consumed by the load, in W, is _____,

Q7.14.1 NAT 2014 2M

The total power dissipated in the circuit, shown in the figure, is 1 kW.

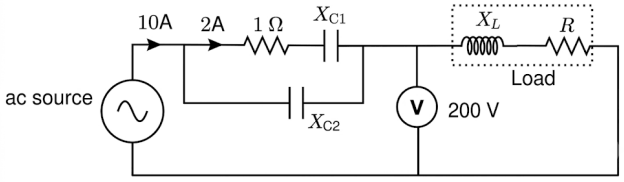


Fig. Q7.14.1

The voltmeter, across the load, reads 200V. The value of X_L is _____.

Q7.13.1 MCQ 2013 1M

A single-phase load is supplied by a single-phase voltage source. If the current flowing from the load to the source is $10\angle -150^\circ$ A and if the voltage at the load terminals is $100\angle 60^\circ$, then the

A. load absorbs real power and delivers reactive power.

B. load absorbs real power and absorbs reactive power.

C. load delivers real power and delivers reactive power.

D. load delivers real power and absorbs reactive power.

Q7.13.2 MCQ 2013 1M

A source $v_s(t) = V\cos 100\pi t$ has an internal impedance of (4 + j3) Ω. If a purely resistive load connected to this source has to extract the maximum power out of the source, its value in Ω should be

A. 3 **B.** 4 **C.** 5 **D.** 7

Q7.12.1 MCQ 2012 1M

The average power delivered to an impedance (4-j3) Ω by a current $5\cos(100\pi t + 100)$ A is

A. 44.2W **B.** 50W **C.** 62.5W **D.** 125W

Q7.12.2 MCQ 2012 2M

Assuming both the voltage sources are in phase, the value of R for which maximum power is transferred from circuit A to circuit B is

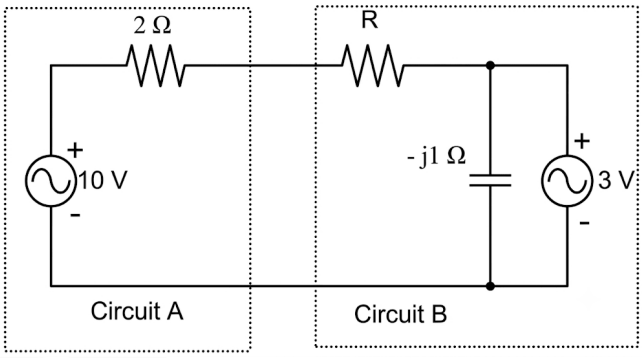


Fig. Q7.12.2

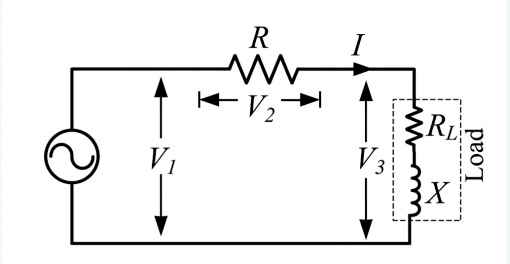
A. 0.8Ω **B.** 1.4Ω **C.** 2Ω **D.** 2.8Ω

Common Data Questions

The next 2 questions share the data below.

Statement for Linked Answer Questions 7.12.3 and 7.12.4:

In the circuit shown, the three voltmeter readings are $V_1 \cong 220V$, $V_2 \cong 122V$, $V_3 \cong 136V$.



Q7.12.3 MCQ 2012 2M

The power factor of the load is

A. 0.45 **B.** 0.50 **C.** 0.55 **D.** 0.60

Q7.12.4 MCQ 2012 2M

If $R_L \cong 5\Omega$, the approximate power consumption in the load is

A. 700W **B.** 750W **C.** 800W **D.** 850W

Q7.04.1 MCQ 2004 1M

Total instantaneous power supplied by a 3-phase ac supply to a balanced R-L load is

A. Zero

B. Constant

C. Pulsating with zero average

D. Pulsating with non-zero average

Q7.01.1 MCQ 2001 2M

A 240 V single-phase ac source is connected to a load with an impedance of $10\angle 60°\Omega$. A capacitor is connected in parallel with the load. If the capacitor supplied 1250 VAR, the real power supplied by the source is

A. 3600 W **B.** 2880 W
C. 2400 W **D.** 1200 W

Q7.99.1 MCQ 1999 2M

The voltage phasor of a circuit is $10\angle 15°V$ and the current phasor is $2\angle -45°A$. The active and the reactive powers in the circuit are :

A. 10 W and 17.32 VAr

B. 5 W and 8.66 VAr

C. 20 W and 60 VAr

D. $20\sqrt{2}$ W and $10\sqrt{2}$ VAr

Q7.96.1 MCQ 1996 1M

A water boiler at home is switched on the a.c. mains supplying power at 230V/ 50Hz. The frequency of instantaneous power consumed by the boiler is

A. 0Hz **B.** 50Hz **C.** 100Hz **D.** 150Hz

Q7.91.1 NAT 1991 2M

In figure calculate

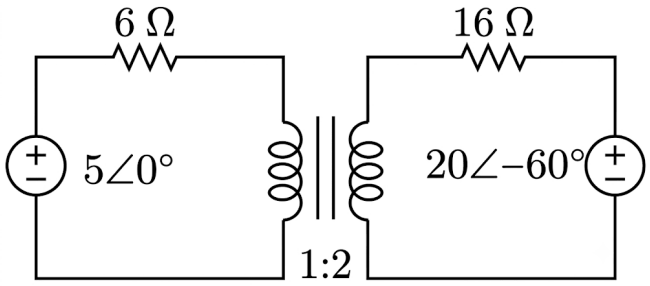


Fig. Q7.91.1

(a) The power delivered by each source
(b) The power dissipated in each resistor

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q7.26.1 **B**	Q7.26.2 **C**	Q7.25.1 **0.707**
Q7.19.1 **585.0 to 590.0**	Q7.18.1 **C**	Q7.18.2 **10.0 to 10.0**
Q7.17.1 **A**	Q7.16.1 **249.0 to 251.0**	Q7.14.1 **17.3 to 17.4**
Q7.13.1 **B**	Q7.13.2 **C**	Q7.12.1 **B**
Q7.12.2 **A**	Q7.12.3 **A**	Q7.12.4 **B**
Q7.04.1 **B**	Q7.01.1 **B**	Q7.99.1 **A**
Q7.96.1 **C**	Q7.91.1 *	

UNIT

VIII

Two Port Parameters

Topics

1. Z-Parameters
2. Y-Parameters, Reciprocity and Symmetry
3. H, G and ABCD Parameters
4. Series and Parallel Connection of Two Port Networks

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 19	**MCQ:** 11	**MSQ:** 0	**NAT:** 7
1M: 6	**2M:** 13	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2018)–2M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Q8.26.1 NAT 2026 2M

Consider the two-port network shown. For maximum power transfer to the resistive load (R_L), the value of R_L should be _____ Ω. *(Round off to two decimal places)*

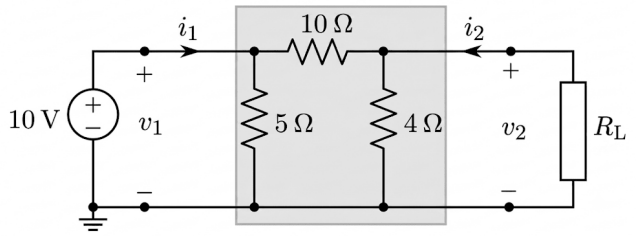


Fig. Q8.26.1

Q8.24.1 NAT 2024 2M

Two passive two-port networks **P** and **Q** are connected as shown in the figure. The impedance matrix of network **P** is $\mathbf{Z_P} = \begin{bmatrix} 40\,\Omega & 60\,\Omega \\ 80\,\Omega & 100\,\Omega \end{bmatrix}$. The admittance matrix of network **Q** is $\mathbf{Y_Q} = \begin{bmatrix} 5\,\text{S} & -2.5\,\text{S} \\ -2.5\,\text{S} & 1\,\text{S} \end{bmatrix}$. Let the ABCD matrix of the two-port network **R** in the figure be $\begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}$. The value of β in Ω is _____ (rounded off to 2 decimal places).

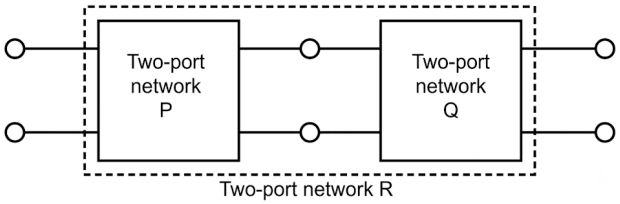


Fig. Q8.24.1

Q8.23.1 NAT 2023 2M

The admittance parameters of the passive resistive two-port network shown in the figure are:
$y_{11} = 5\,\text{S}, y_{22} = 1\,\text{S}, y_{12} = y_{21} = -2.5\,\text{S}$
The power delivered to the load resistor R_L in Watt is _____ (Round off to 2 decimal places).

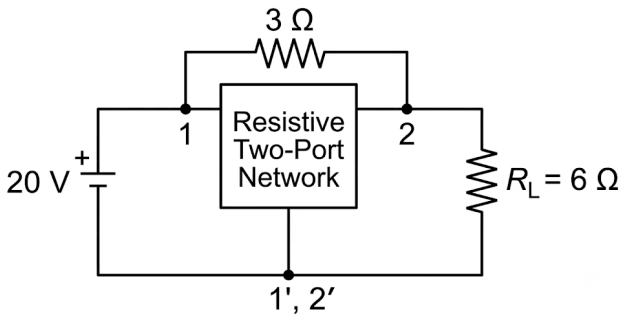


Fig. Q8.23.1

Q8.18.1 NAT 2018 1M

In the two-port network shown, the h_{11} parameter (where, $h_{11} = \frac{V_1}{I_1}$ when $V_2 = 0$) in ohms is _____ (up to 2 decimal places).

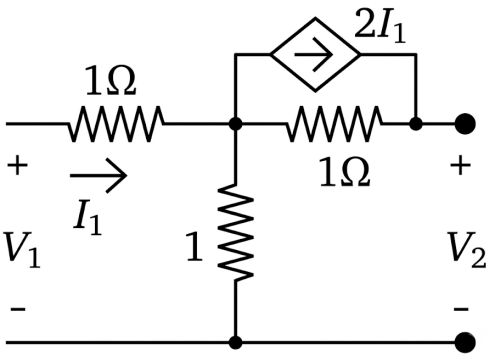


Fig. Q8.18.1

Q8.17.1 NAT 2017 1M

For the given two port network, the value of transfer impedance z_{21} in ohms is _____.

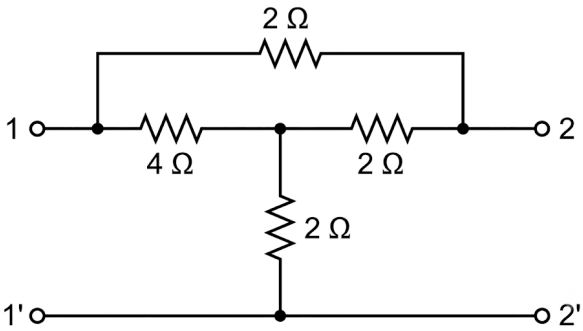


Fig. Q8.17.1

Q8.17.2 MCQ 2017 2M

Two passive two-port networks are connected in cascade as shown in figure. A voltage source is connected at port 1.

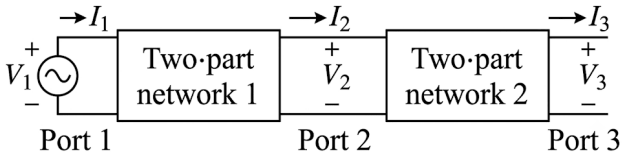


Fig. Q8.17.2

Given, $V_1 = A_1V_2 + B_1I_2$
$I_1 = C_1V_2 + D_1I_2$
$V_2 = A_2V_3 + B_2I_3$
$I_2 = C_2V_3 + D_2I_3$

$A_1, B_1, C_1, D_1, A_2, B_2, C_2$, and D_2 are the generalized circuit constants. If the Thevenin equivalent circuit at port 3 consists of a voltage source V_T and impedance Z_T, connected in series, then

A. $V_T = \dfrac{V_1}{A_1A_2}, Z_T = \dfrac{A_1B_2 + B_1D_2}{A_1A_2 + B_1C_2}$

B. $V_T = \dfrac{V_1}{A_1A_2 + B_1C_2}, Z_T = \dfrac{A_1B_2 + B_1D_2}{A_1A_2}$

C. $V_T = \dfrac{V_1}{A_1 + A_2}, Z_T = \dfrac{A_1B_2 + B_1D_2}{A_1 + A_2}$

D. $V_T = \dfrac{V_1}{A_1A_2 + B_1C_2}, Z_T = \dfrac{A_1B_2 + B_1D_2}{A_1A_2 + B_1C_2}$

Q8.16.1 NAT 2016 2M

The z-parameters of the two port network shown in the figure are $z_{11} = 40\,\Omega$, $z_{12} = 60\,\Omega$, $z_{21} = 80\,\Omega$, and $z_{22} = 100\,\Omega$. The average power delivered to $R_L = 20\,\Omega$, in watts, is _____.

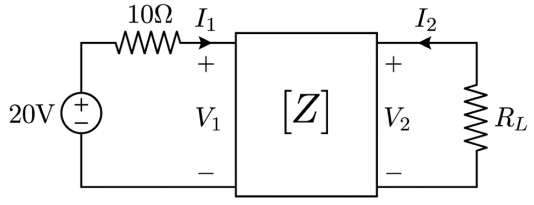


Fig. Q8.16.1

Q8.15.1 NAT 2015 2M

In a linear two-port network, when 10 V is applied to port 1, a current of 4 A flows through port 2 when it is short – circuited. When 5 V is applied to port 1, a current of 1.25 A flows through a 1 Ω resistance connected across port 2. When 3 V is applied to port 1, the current (in Ampere) through a 2 Ω resistance connected across port 2 is _____.

Q8.10.1 MCQ 2010 2M

The two-port network **P** shown in the figure has ports 1 and 2, denoted by terminals (a, b) and (c, d) respectively. It has an impedance matrix **Z** with parameters denoted by Z_{ij}. A 1 Ω resistor is connected in series with the network at port 1 as shown in the figure. The impedance matrix of the modified two-port network (shown as a dashed box) is

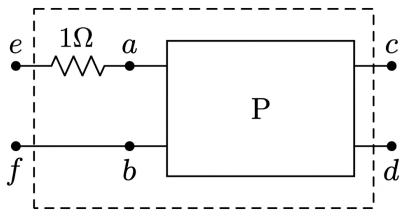


Fig. Q8.10.1

A. $\begin{pmatrix} Z_{11}+1 & Z_{12}+1 \\ Z_{21} & Z_{22}+1 \end{pmatrix}$ **B.** $\begin{pmatrix} Z_{11}+1 & Z_{12} \\ Z_{21} & Z_{22}+1 \end{pmatrix}$

C. $\begin{pmatrix} Z_{11}+1 & Z_{12} \\ Z_{21} & Z_{22} \end{pmatrix}$ **D.** $\begin{pmatrix} Z_{11}+1 & Z_{12} \\ Z_{21}+1 & Z_{22} \end{pmatrix}$

Q8.06.1 MCQ 2006 2M

The parameter type and the matrix representation of the relevant two port parameters that describe the circuit shown are

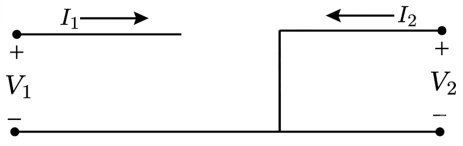


Fig. Q8.06.1

A. z parameters, $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ **B.** h parameters, $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

C. h parameters, $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ **D.** z parameters, $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Q8.05.1 MCQ 2005 2M

Two networks are connected in cascade as shown in Figure. With the usual notations the equivalent A, B, C and D constants are obtained. Given that $C = 0.025\angle 45°$, the value of Z_2 is:

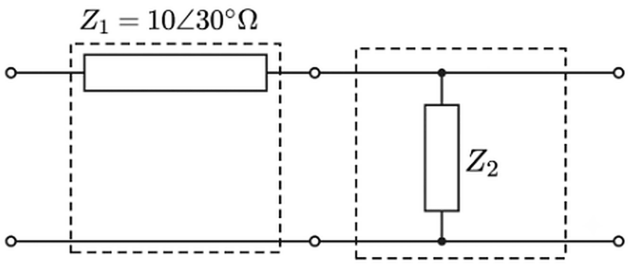


Fig. Q8.05.1

A. $10\angle 30°\ \Omega$ **B.** $40\angle -45°\ \Omega$
C. $1\ \Omega$ **D.** $0\ \Omega$

Q8.05.2 MCQ 2005 1M

For the two-port network shown in Figure, the Z-matrix is given by

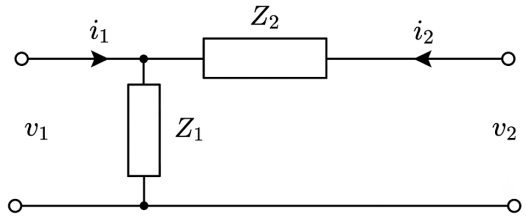


Fig. Q8.05.2

A. $\begin{bmatrix} Z_1 & Z_1+Z_2 \\ Z_1+Z_2 & Z_2 \end{bmatrix}$ **B.** $\begin{bmatrix} Z_1 & Z_1 \\ Z_1+Z_2 & Z_2 \end{bmatrix}$

C. $\begin{bmatrix} Z_1 & Z_2 \\ Z_1 & Z_1+Z_2 \end{bmatrix}$ **D.** $\begin{bmatrix} Z_1 & Z_1 \\ Z_1 & Z_1+Z_2 \end{bmatrix}$

Q8.03.1 MCQ 2003 2M

The h-parameters for a two-port network are defined by

$$\begin{bmatrix} E_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ E_2 \end{bmatrix}$$

For the two-port network shown in Figure, the value of h_{12} is given by

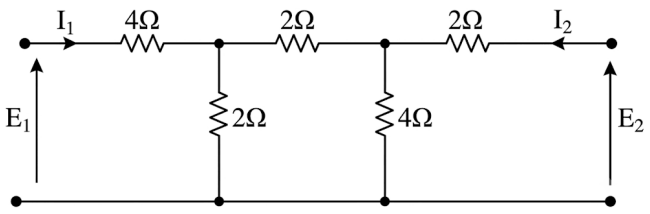


Fig. Q8.03.1

A. 0.125 **B.** 0.167 **C.** 0.625 **D.** 0.25

Q8.02.1 MCQ 2002 2M

A two port network, shown in Figure is described by the following equations.

$$I_1 = Y_{11}E_1 + Y_{12}E_2$$

$$I_2 = Y_{21}E_1 + Y_{22}E_2$$

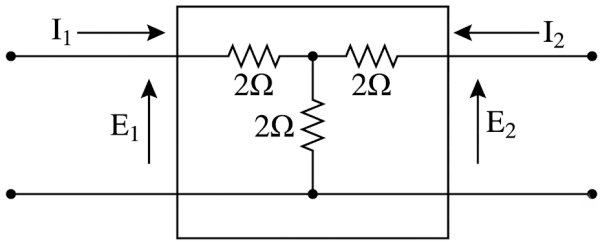


Fig. Q8.02.1

The admittance parameters Y_{11}, Y_{12}, Y_{21}, Y_{22} for the network shown are

A. 0.5 mho, 1 mho, 2 mho and 1 mho respectively

B. $\frac{1}{3}$ mho, $-\frac{1}{6}$ mho, $-\frac{1}{6}$ mho and $\frac{1}{3}$ mho respectively

C. 0.5 mho, 0.5 mho, 1.5 mho and 2 mho respectively

D. $-\frac{2}{5}$ mho, $-\frac{3}{7}$ mho, $\frac{3}{7}$ mho and $\frac{2}{5}$ mho respectively

Q8.01.1 MCQ 2001 1M

A passive 2-port network is in a steady-state. Compared to its input, the steady state output can never offer

A. Higher voltage **B.** Lower impedance
C. Greater power **D.** Better regulation

Q8.00.1 MCQ 2000 2M

A two-port device is defined by the following pair of equations: $i_1 = 2v_1 + v_2$ and $i_2 = v_1 + v_2$
Its impedance parameters $z_{11}, z_{12}, z_{21}, z_{22}$ are given by

A. $(2, 1, 1, 1)$ **B.** $(1, -1, -1, 2)$
C. $(1, 1, 1, 2)$ **D.** $(2, -1, -1, 1)$

Q8.96.1 Descriptive 1996 2M

The performance of a general two-port network N shown in the figure can be described in terms of its z-parameters or h-parameters as indicated below

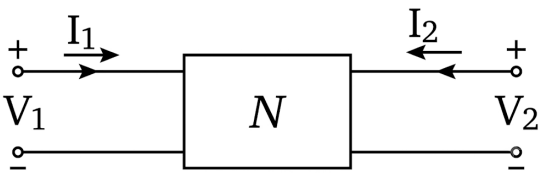


Fig. Q8.96.1

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

If the two-port is further characterized by additional constraints, a few conditions have to be satisfied by the parameters of the network. For each of the constraints specified in Table A, choose the appropriate condition to be satisfied, from amongst those listed in Table B.

Table A	Table B
(a) Reciprocity at the ports	(P) $h_{11}h_{22} - h_{12}h_{21} = 1$
(b) Passivity at the ports	(Q) $z_{12} = -z_{21}$
(c) Electrical symmetry at the ports	(R) $h_{12} = h_{21}$
	(T) absence of negative resistors inside the 2-port
	(U) energy input to the network from the ports is non-negative for all conditions at the ports

Q8.94.1 MCQ 1994 1M

If a two-port network is passive, then we have, with the usual notation, the following relationship

A. $h_{12} = h_{21}$

B. $h_{12} = -h_{21}$

C. $h_{11} = h_{22}$

D. $h_{11} \cdot h_{22} - h_{12} \cdot h_{21} = 1$

Q8.91.1 MCQ 1991 1M

The terminal voltage and currents of a two-port network are indicated on the figure. If the two-port is reciprocal, then

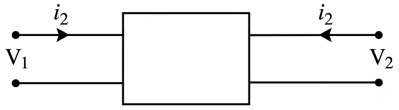


Fig. Q8.91.1

A. $\dfrac{Z_{12}}{Y_{12}} = Z_{12}^2 - Z_{11} \cdot Z_{22}$

B. $Z_{12} = \dfrac{1}{Y_{22}}$

C. $h_{12} = -h_{21}$

D. $AD - BC = 0$

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q8.26.1 2.81 to 2.91	Q8.24.1 -19.90 to -19.70	Q8.23.1 237 to 239
Q8.18.1 0.5	Q8.17.1 3	Q8.17.2 D
Q8.16.1 35.55	Q8.15.1 0.4 to 0.6	Q8.10.1 C
Q8.06.1 *	Q8.05.1 B	Q8.05.2 D
Q8.03.1 D	Q8.02.1 B	Q8.01.1 C
Q8.00.1 B	Q8.96.1 *	Q8.94.1 B
Q8.91.1 C		

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SiliconX	Electronic Devices, Electromagnetic Field Theory and Communication Systems — EC
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Engineering Mathematics	Linear Algebra, Calculus, Differential Equations, Complex Variables, Probability & Statistics, Numerical Methods, Transform Theory and Vector Calculus — all branches
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